The Real Cost of Benchmarking

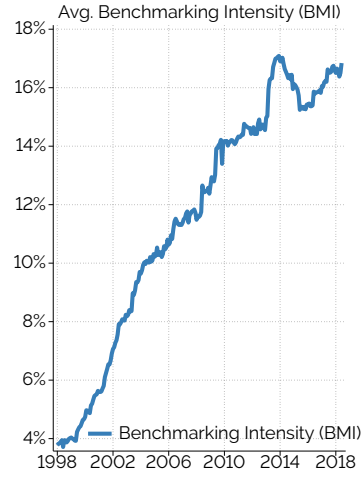
Christian Kontz
Stanford GSB

Sebastian Hanson
Stanford GSB

January 2026

Does benchmark-linked ownership affect the real economy?

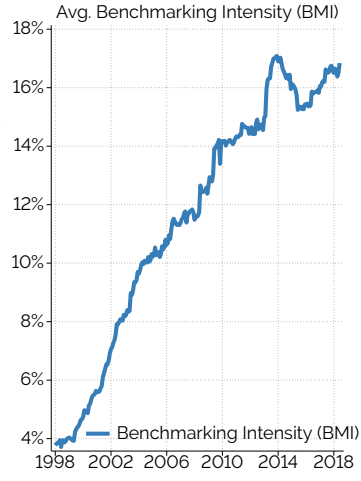
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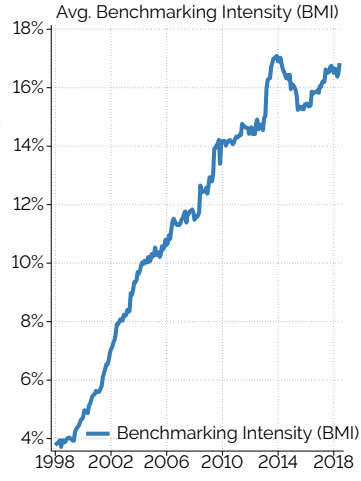
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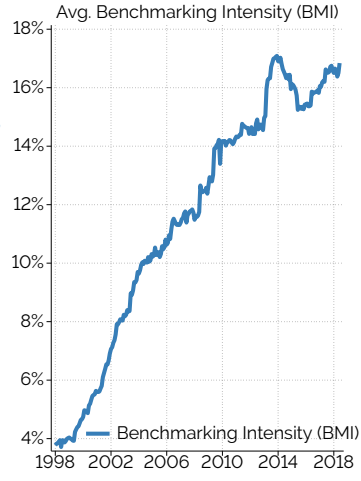


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This paper:

BMI $\uparrow \rightarrow$ **CAPM** $\hat{\beta} \uparrow \rightarrow$ **CAPX** $\searrow$ **and payouts** $\nearrow$



Preview: Benchmarking raises perceived cost of equity and thus lowers investment

1. Benchmark-linked ownership and passive investing changed **cross-section of CAPM $\hat{\beta}$s**
   - Equal-weighted average β increased by 0.3 since 1990s $\rightarrow$ 180 bps increase in cost of equity
   - BMI $\times$ Passive flow factor explains $\hat{\beta}$s in cross-section and time series

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4. Cross-sectional **changes in investment do not cancel** even though $\beta^{\textit{Mkt}} := \sum \omega \hat{\beta} = 1$
   - Elasticity of investment wrt cost of capital not uniform $\rightarrow$ distribution of $\hat{\beta}$ matters

Contribution to the Literature

Excess Comovement of Stock Returns

- Pindyck & Rotemberg (1993), Campbell & Mei (1993), **Vijh (1994)**, Barberis & Shleifer (2003), Barberis, Shleifer, Wurgler (2005), Kumar & Lee (2005), Greenwood (2008), Greenwood & Thesmar (2011), Boyer (2011), Antón & Polk (2014), **Buffa & Hodor (2022)**, **Fang et al. (2024)**

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(Real-) Effects of Benchmark-linked and Passive Index Investing

- Brennan (1993), Admati & Pfleiderer (1997), Basak & Pavlova (2013), Bena et al. (2017), Brogaard et al. (2019), Coles et al. (2019), Appel et al. (2019), Cremers et al. (2020), Kacperczyk et al. (2021), Bond & Garcia (2022), **Kashyap, Kovrijnykh, Li, Pavlova (2021)**, Pavlova & Sikorskaya (2023), Bennett et al. (2023), Antoniou et al. (2023), **Buss & Sundaresan (2023)**, Sikorskaya (2024), Chaudhry (2025), Sharma (2026)

➠ **This paper: Benchmarking *lowers* investment by increasing perceived cost of equity**

Plan for today's talk

1. MBA Textbook Model and **Data**
2. Three Novel Stylized Facts About the **Cross-section of Benchmark Stocks**
3. Effects of Benchmark-linked Ownership on CAPM β and **Perceived Cost of Equity**
4. Effects of Benchmark-linked Ownership on **Firm Investment**
5. Effects of Benchmark-linked Ownership on **Aggregate Investment**

MBA Textbook Model and Data

MBA textbook: CAPM β determines firm value and investment

- Manager maximizes firm value using **CAPM-implied discount rate:** $R = R^f + \beta \times 6\%$
- Manager sets **investment policy** using NPV rule: $\mathbb{E}\left[\frac{\text{Net Cashflow}}{R^f+\beta\times 6\%}\right] > 0$

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- Inelastic demand of **benchmark-linked ownership** affects discount rate

$$\widetilde{R} = R^f + [\beta + \Delta_\beta(BMI)] \times 6\% - \Delta_P(BMI)$$

- Benchmarking has **two competing effects** that increase in benchmarking intensity
  1. Price effect $\Delta_P(BMI) \rightarrow$ lowers discount rate $\rightarrow$ more investment
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NB: Managers using CAPM observe $\Delta_\beta(BMI)$ but won't internalize $\Delta_P(BMI)$

▸ Micro foundations for wedges ▸ BMI vs. estimates of demand elasticities from HHL (2025, AER) ▸ Literature: managers use CAPM

Our sample covers U.S. common stock from 1998 to 2018

1. Benchmarking intensity from Pavlova & Sikorskaya (2023)

$$\textbf{BMI}_i = \sum_j \mathbb{1}\{\text{stock } i \in \text{index } j\} \times \frac{\text{AUM benchmarked to index } j}{\text{market capitalization of index } j}$$

➟ **Measure of inelastic demand by benchmarked mutual funds and ETFs**
- Captures 34 U.S. equity indices (S&P, Russell, MSCI, CRSP, Dow Jones, FTSE, Wilshire, ...)
- Representing 90% of AUM in U.S. equity mutual funds and ETFs

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2. CAPM $\hat{\beta}$ with respect to CRSP universe

3. Perceived cost of equity from 6 different sources
   - Firm managers (Gormsen & Huber, 2024)
   - Stock analysts (Morningstar, Value Line, I/B/E/S)
   - Regulators (utilities, railroads)

4. Firm- and industry-level variables from standard sources (e.g., S&P's Compustat)

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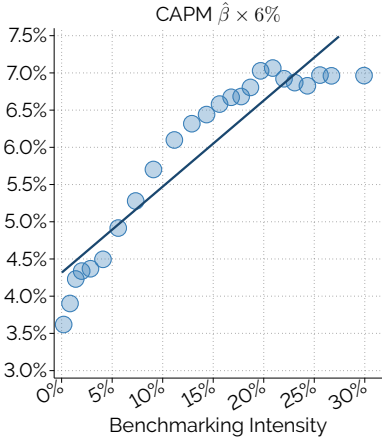
➟ **Limitation**: no direct information about the actual CAPM $\hat{\beta}$ used by managers

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Three Novel Stylized Facts About Benchmark Stocks

Fact 1: Benchmarking intensity positively correlates with CAPM β estimates

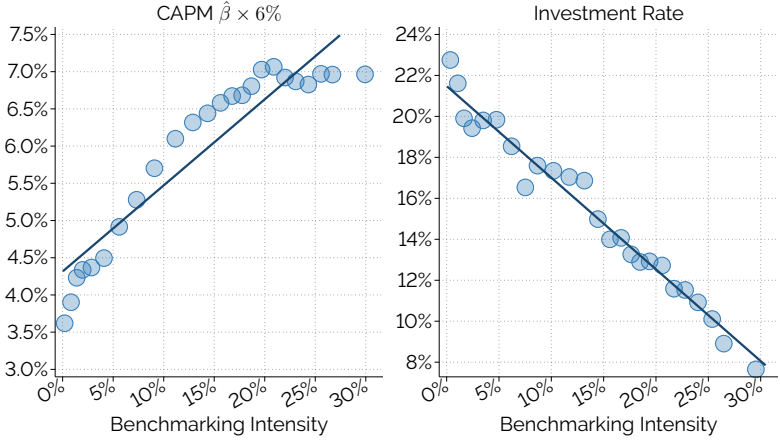
$$\text{CAPM } \hat{\beta} \times 6\% = \textbf{BMI} + \text{Year} \times \text{Size FE} + \varepsilon$$



▸ Fact 1 vs ranks ▸ Other β freq. (reg) ▸ Variation within cell

Fact 2: Benchmarking intensity negatively correlates with firm investment

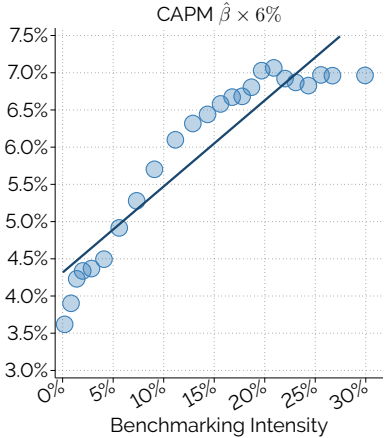
$$\text{Investment Rate} = \textbf{BMI} + \text{Year} \times \text{Size FE} + \varepsilon$$

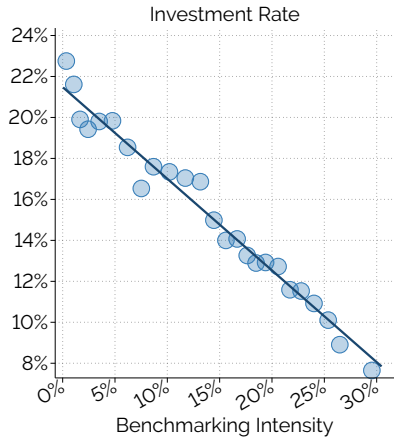


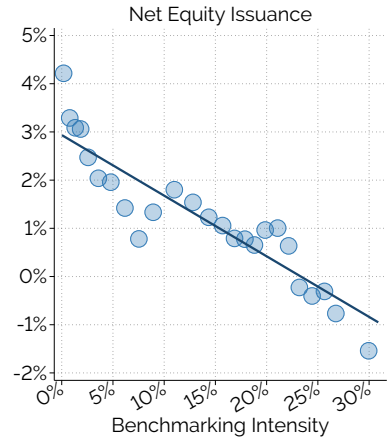
▸ Fact 1 vs ranks ▸ Other β freq. (reg) ▸ Variation within cell ▸ Fact 2 vs ranks ▸ Other inv. rate defs.

Fact 3: Benchmarking intensity negatively correlates with equity issuance

$$\text{Net Equity Issuance} = \textbf{BMI} + \text{Year} \times \text{Size FE} + \varepsilon$$







▸ Fact 1 vs ranks ▸ Other β freq. (reg) ▸ Variation within cell ▸ Fact 2 vs ranks ▸ Other inv. rate defs. ▸ Fact 3 vs ranks ▸ HHL (2025)

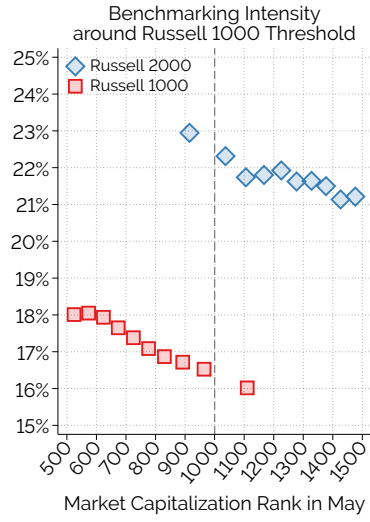
Effects of Benchmark-linked Ownership on CAPM $\hat{\beta}s$

Identification strategy uses exog. variation in BMI from Russell reconstitution

- Around **27% of AUM** in U.S. domestic equity funds are **benchmarked to Russell** indices (Pavlova & Sikorskaya, 2023)

- Each year Russell ranks all common stocks by mkt cap
  - Top 1000 → Russell 1000
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NB: **Russell 2000** is a popular **benchmark** for **passive index funds** ▸ IOR



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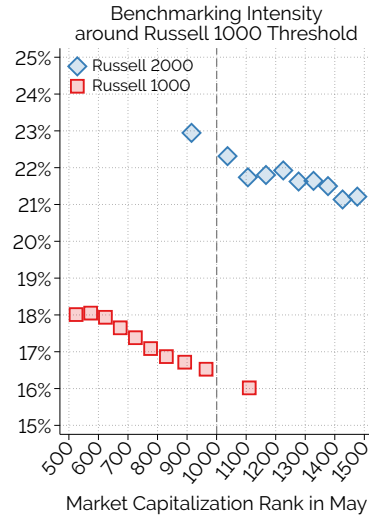
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- We account for particularities of Russell identification
  - Ranking variable (proxy of Ben-David et al., 2019)
  - Banding variables post-2007 (Appel et al., 2019)
  - Liquidity, bid-ask spreads (Pavlova & Sikorskaya, 2023)
  - Window of +/- 300 ranks around threshold

- Identification assumption: firms do not self-select into index

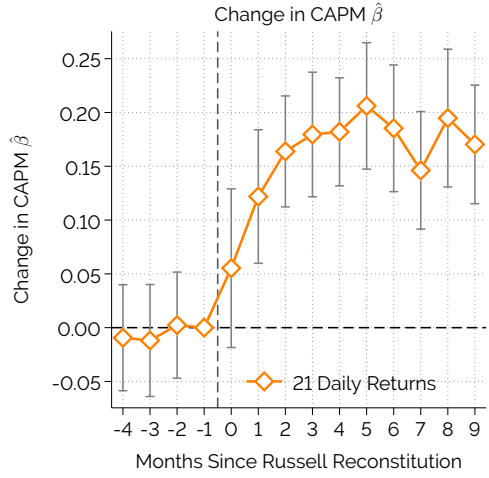


▸ Accuracy of rank proxy ▸ Relevance condition of ΔBMI ▸ Replication of Pavlova & Sikorskaya (2023) ▸ Exclusion restriction tests

Exogenous increase in BMI leads to increase in CAPM $\hat{\beta}$

$$\text{CAPM } \hat{\beta}^{21d} = \sum_{k=-4}^{9} \gamma_k \left(\Delta \text{BMI} \times \mathbb{1}\{t \geq \textit{June}\}\right) + \text{Ind.} \times \text{Time FE} + \text{Stock FE} + \text{Controls} + \varepsilon$$

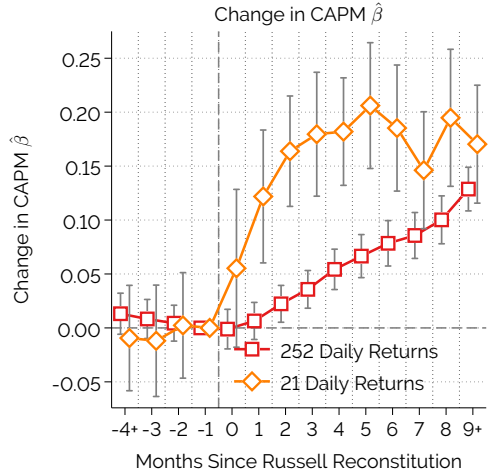
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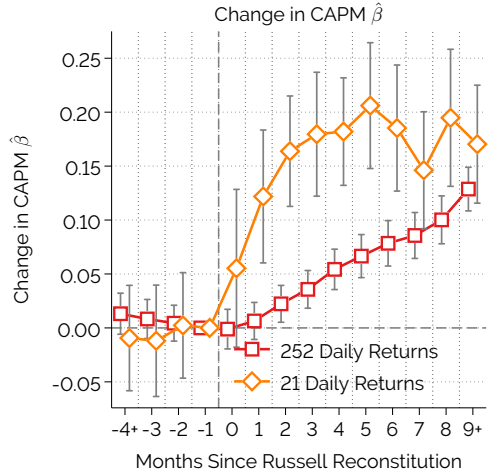


▸ Which β do managers use? ▸ Fact 1 ▸ Ew. $\bar{\beta}$ ▸ Rank proxy ▸ Relevance ▸ S&P 500 $\hat{\beta}$ ▸ Liquidity ▸ Pre-treatment β ▸ Price effect

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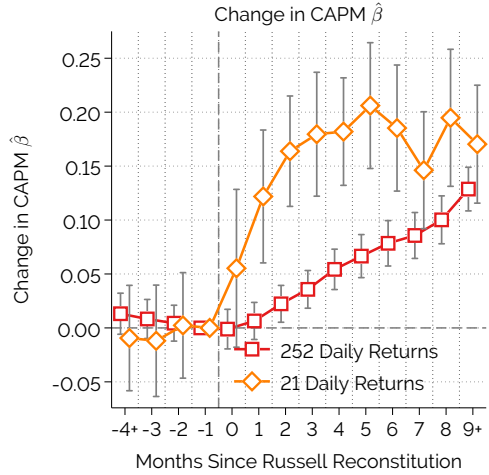


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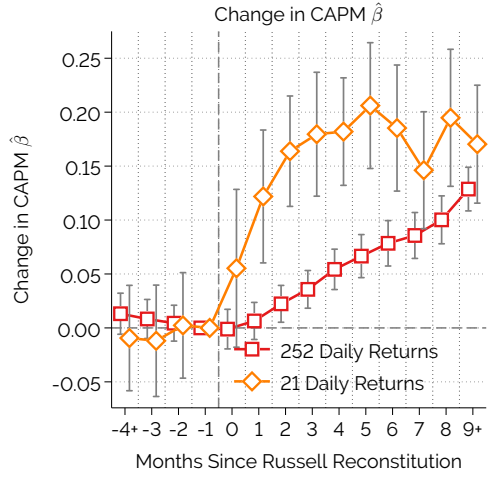


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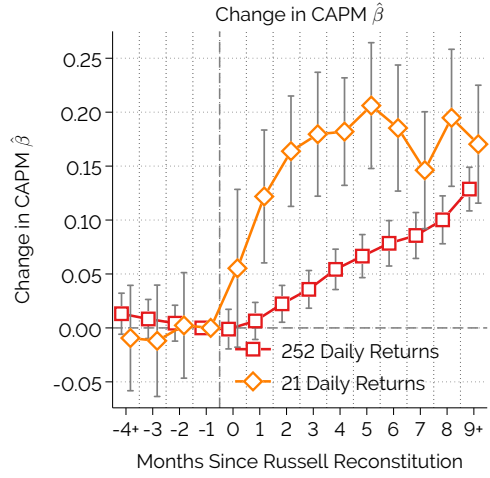


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- **Flows** into **passive funds and ETFs** are key driver



▸ Which β do managers use? ▸ Fact 1 ▸ Ew. $\bar{\beta}$ ▸ Rank proxy ▸ Relevance ▸ S&P 500 $\hat{\beta}$ ▸ Liquidity ▸ Pre-treatment β
▸ Price effect ▸ Persist. shock ▸ Within-R^2 ▸ Shapley ▸ Fundamentals ▸ Freq./Price pressure ▸ Flows ▸ Simulation

Effects of Benchmark-linked Ownership on Perceived Cost of Equity

Firm managers use CAPM to estimate cost of equity in practice

Surveys

- Graham & Harvey (2001): 74% of CFOs (almost) always use CAPM
- Jacobs & Shivdasan (2012): 90% use CAPM in DCF
- Mukhlynina & Nyborg (2016): 76% (almost) always use CAPM
- Graham (2022): 80% of CFOs (almost) always use CAPM

Revealed Preferences

- Krüger et al. (2015): managers use CAPM in capital budgeting
- Jaganathan et al. (2016): CAPM βs explain discount rates of CFOs
- Dessaint et al. (2020): managers use CAPM to value acquisitions
- Gormsen & Huber (2024): perc. cost of capital correlates with CAPM
- Cho & Salarkia (2025): managers use CAPM to time repurchases

Other Professions

- Berk & Binsbergen (2015); Barber et al. (2016): investors use CAPM
- Jensen (2023): stock analysts use CAPM
- Décaire & Graham (2025): 96.8% of stock analyst reports use CAPM
- Kontz (2025): utility regulators use CAPM

▸ More CAPM quotes ▸ Exec. comments on Russell index inclusion

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*"Why do we think that a 10% return is good? Well, you have to — whether we are creating **shareholder value** really goes to what's our **cost of capital**. […]*
*This is really how we view our WACC. Most of you will bring back **visions of business school**. This is the **capital asset pricing model**, right? Our **cost of equity**, about 10.7%."*

— CFO, LKQ Corp. (Q1 2016)

▸ More CAPM quotes ▸ Exec. comments on Russell index inclusion

$$\Delta\text{Perc. CoC} = \alpha_t + \gamma\, \Delta\text{BMI} + \text{Controls} + \nu \qquad \text{(RF)}$$

$$\Delta\text{CAPM } \hat{\beta} = \alpha_t + \pi\, \Delta\text{BMI} + \text{Controls} + \varepsilon \qquad \text{(FS)}$$

$$\Delta\text{Perc. CoC} = \alpha_t + \lambda\, \widehat{\Delta\text{CAPM } \beta} + \text{Controls} + \epsilon \qquad \text{(IV)}$$

- $\mathbb{E}^{\star}[\text{equity risk premium}] \quad \widehat{\lambda} = \widehat{\gamma} \,/\, \widehat{\pi}$

▸ Details Data

Increases in BMI raise perceived cost of equity in 7 data sets

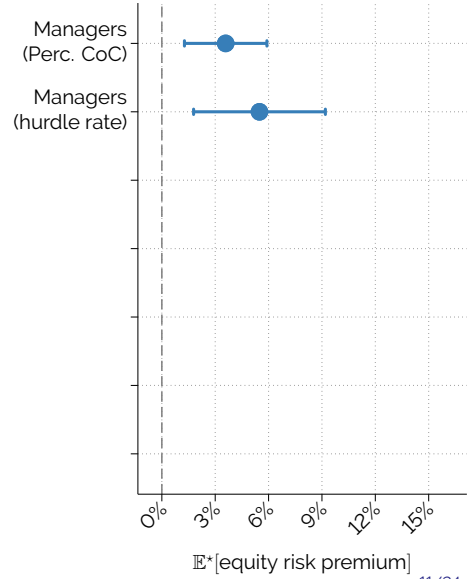
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- $\mathbb{E}^{\star}$[equity risk premium] $\widehat{\lambda} = \widehat{\gamma}\ /\ \widehat{\pi}$
  - Firm managers $\approx 4.5\%$

▸ Details Data ▸ Table Managers



Increases in BMI raise perceived cost of equity in 7 data sets

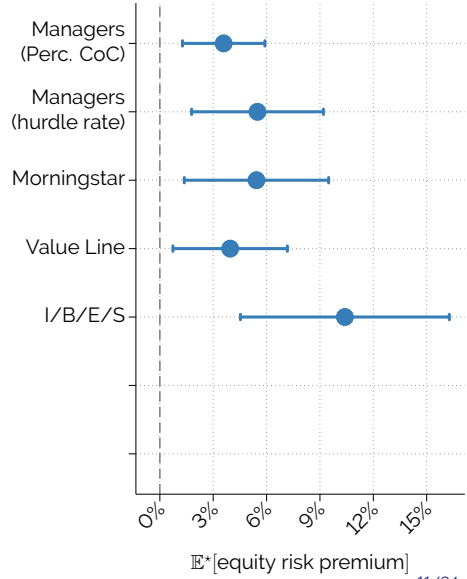
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- $\mathbb{E}^*$[equity risk premium] $\widehat{\lambda} = \widehat{\gamma}\ /\ \widehat{\pi}$
  - Firm managers $\approx 4.5\%$
  - Stock analysts $\approx 6.6\%$

▸ Details Data ▸ Table Managers ▸ Table Analysts



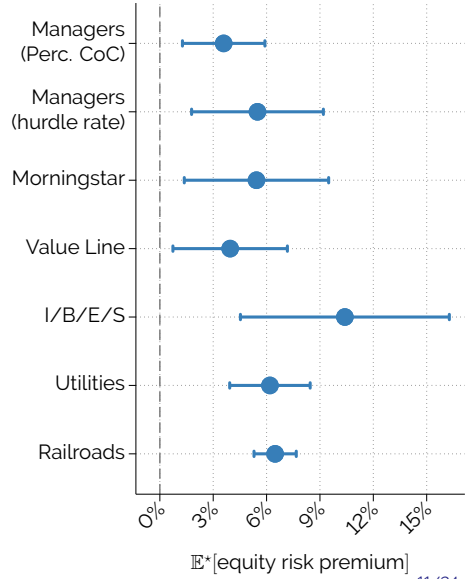
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$$\Delta\text{Perc. CoC} = \alpha_t + \lambda\ \widehat{\Delta\text{CAPM}\ \beta} + \text{Controls} + \epsilon$$

- $\mathbb{E}^\star$[equity risk premium] $\widehat{\lambda} = \widehat{\gamma}\ /\ \widehat{\pi}$
  - Firm managers $\approx 4.5\%$
  - Stock analysts $\approx 6.6\%$
  - Regulators $\approx 6.3\%$



▸ Details Data ▸ Table Managers ▸ Table Analysts ▸ Table Regulators
▸ Placebo test with regulatory cost of debt ▸ Price effect ▸ Est. in Diffs.
▸ PCoC passthru to hurdle rate ▸ Which β do managers use?

Effects of Benchmark-linked Ownership on Investment

$$Inv_{i,t+1} = \text{BMI}_{i,t} + \text{Tobin's } q_{i,t}^{tot} + \text{Firm FE} + \text{Year FE} + \text{Controls} + \varepsilon_{i,t+1}$$

1 SD higher BMI is associated with a 13% *lower* investment rate

$$Inv_{i,t+1} = \text{BMI}_{i,t} + \text{Tobin's } q^{tot}_{i,t} + \text{Firm FE} + \text{Year FE} + \text{Controls} + \varepsilon_{i,t+1}$$

	Dep. var.: $Inv_{i,t+1} = \text{CAPX}_{t+1}$ / PPE_t (in %)				
BMI (in %)	-0.233***	-0.300***	-0.205***	-0.111***	-0.130***
	(0.051)	(0.038)	(0.041)	(0.023)	(0.029)
Linear Time Trend	-0.433***				
	(0.078)				
Tobin's q^{tot}				2.881***	2.814***
				(0.186)	(0.182)
Firm FE	✓		✓	✓	✓
Year × Size FE		✓	✓	✓	
Year × Size × Ind. FE					✓
Other Controls				✓	✓
Adj. R^2	0.42	0.08	0.45	0.51	0.63
Mean Dep. Var.	15.0	15.0	15.0	15.0	15.0
Std. dev. BMI	9.4	9.4	9.4	9.3	9.2
Observations	36,787	36,843	36,782	35,181	34,574

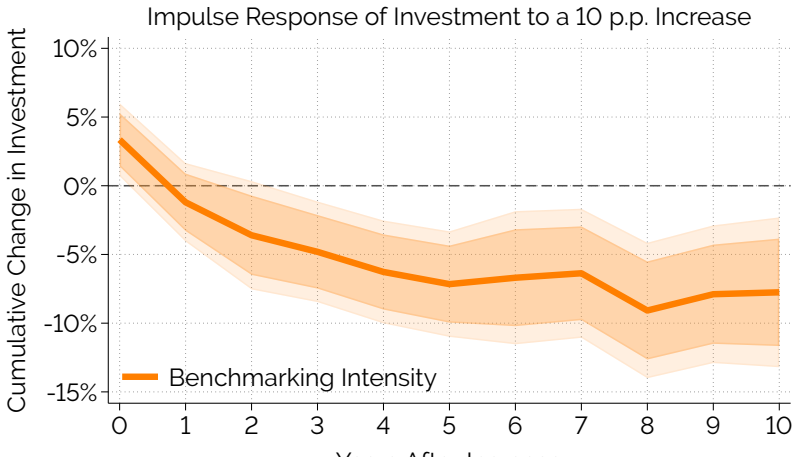
▸ Adjusting Tobin's q for measurement error ▸ Intangible capital investment rate ▸ Interaction Term

Impulse response of investment to year-on-year BMI increase at $h = 0, \ldots, 10$

$$\log\left(Inv_{i,t+h}\right) - \log\left(Inv_{i,t-1}\right) = \gamma^h(\text{BMI}_{i,t} - \text{BMI}_{i,t-1}) + \text{Ind.} \times \text{Year FE} + \text{Controls} + \varepsilon_{i,t+h}$$

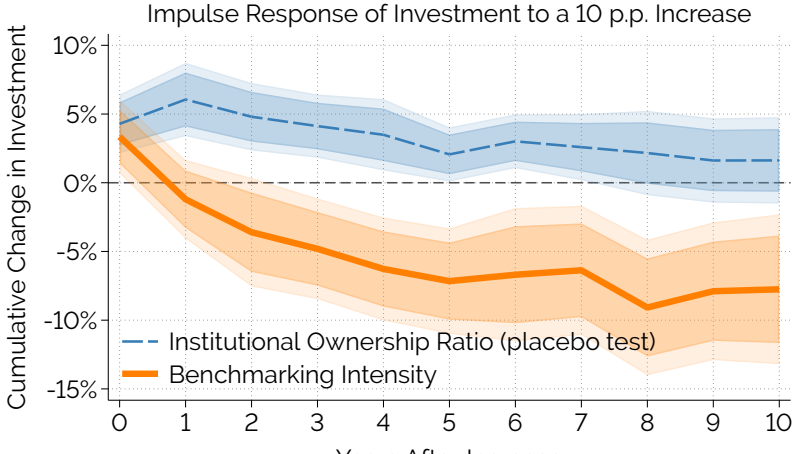
Investment falls by 7.8% in response to a 10 p.p. year-on-year BMI increase

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$$= \theta^h(\text{IOR}_{i,t} - \text{IOR}_{i,t-1}) + \text{Ind.} \times \text{Year FE} + \text{Controls} + \epsilon_{i,t+h}$$

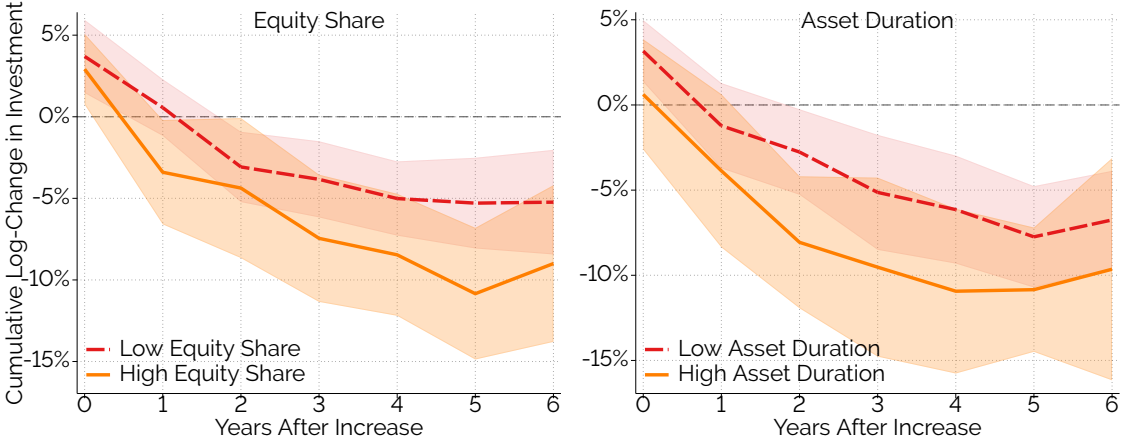


▸ User cost elas. ▸ Intangible inv. ▸ IOR doesn't explain $\hat{\beta}$ ▸ High equity share ▸ Long duration firms

Effects are larger for firms with high equity share or long duration

back

$$\log\left(Inv_{i,t+h}\right) - \log\left(Inv_{i,t-1}\right) = \gamma_1^h \left(\text{BMI}_{i,t} - \text{BMI}_{i,t-1}\right) + \gamma_2^h \, \mathbb{1}\{\text{High}\}_{i,t} \times \left(\text{BMI}_{i,t} - \text{BMI}_{i,t-1}\right) + \mathbb{1}\{\text{High}\}_{i,t} + \text{Ind.} \times \text{Year FE} + \text{Controls} + \varepsilon_{i,t+h}$$

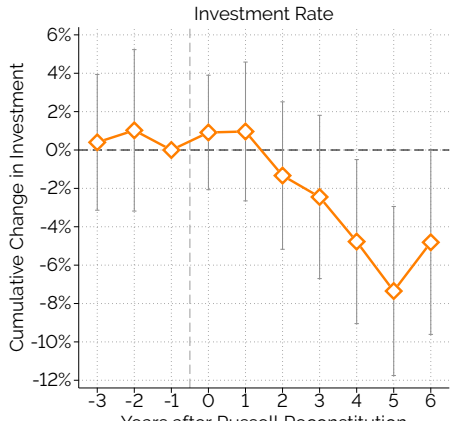


Natural Experiment 1: exog. variation in BMI from Russell Reconstitutions

$$\log\left(Inv_{i,t+h}\right) - \log\left(Inv_{i,t-1}\right) = \gamma^h \Delta \text{BMI}_{i,t} + \text{Ind.} \times \text{Year FE} + \text{Controls} + \varepsilon_{i,t+h}$$

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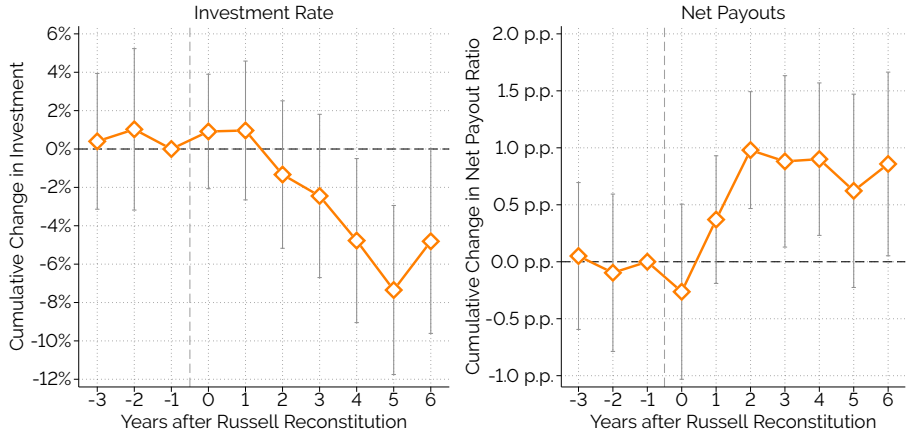
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▸ Table RF ▸ Robustness ▸ Fundamentals ▸ Profitability ▸ Diff. windows ▸ Momentum ▸ R1000/R2000 ▸ Intangibles
▸ Acquisitions ▸ Firms w/ high equity share ▸ Firms w/ capital efficiency targets ▸ Capital Structure ▸ Other inv. rate definitions
▸ Reconciliation with PS23 ▸ R3000 cutoff ▸ LP-IV ▸ Net Payout Decomposition ▸ Persistence of Transition ▸ Pre vs. Post 03

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- **Robustness**
  - Different window widths around cutoff ▸ Result
  - Accounting for momentum ▸ Result
  - Other investment rate definitions ▸ Result
  - Exploiting time-series variation ▸ Result
  - R1000→R2000 vs. R2000→R1000 ▸ Result
  - Additional controls and different fixed effect levels ▸ Result
  - Firm fundamentals / exclusion restriction ▸ Result
  - Reconciliation with PS23 ▸ Result
  - Lower Russell 3000 cutoff ▸ Result

- **Cross-sectional heterogeneity**
  - Firms with high equity share in WACC ▸ Result
  - Firms with capital efficiency targets ▸ Result

- **Other firm outcomes**
  - Intangible investment ▸ Result
  - Acquisitions ▸ Result
  - Profitability ▸ Result
  - Capital structure ▸ Result

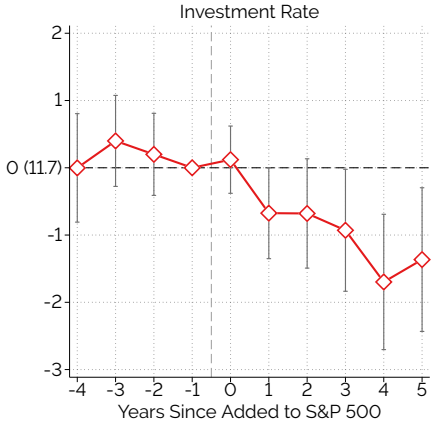
Natural Experiment 2: somewhat less exog. variation in BMI from S&P 500

$$Inv_{i,t} = \sum_{k=-4}^{5} \gamma_k \, \mathbb{1}\{\text{S\&P 500}\} \times \mathbb{1}\{\text{Post}\} + \text{Firm FE} + \text{Ind.} \times \text{Year FE} + \text{Controls} + \varepsilon_{i,t}$$

▸ Change in CAPM $\hat{\beta}$ at inclusion

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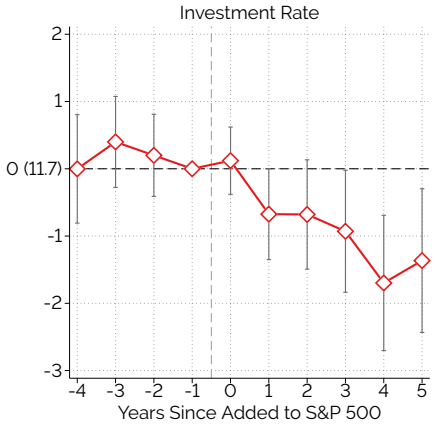
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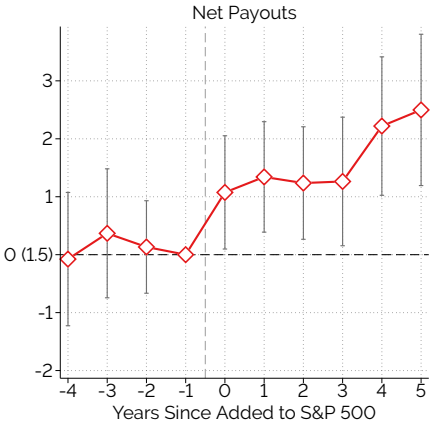


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▸ Change in CAPM $\hat{\beta}$ at inclusion

Main result: BMI $\uparrow \rightarrow$ CAPM $\hat{\beta} \uparrow \rightarrow$ CAPX $\searrow$ and payouts $\nearrow$

Firm-level: natural experiments around benchmark reconstitutions

- Natural Experiment 1: Russell reconstitution using Local Projections
- Natural Experiment 2: S&P 500 addition using Diff-in-Diff

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Industry-level: industries with larger BMI increase accumulated less capital over past 25 years

- Long-difference regressions from 1998 to 2018
- Robust to industry pre-trends and exposure to China shock

$$\log\left(\text{Real Capital}_{i,2018}/\text{Real Capital}_{i,1998}\right) = \alpha_j + \Delta\text{BMI}_i + \varepsilon_i$$

Main result: BMI ↑ → CAPM $\hat{\beta}$ ↑ → CAPX ↘ and payouts ↗

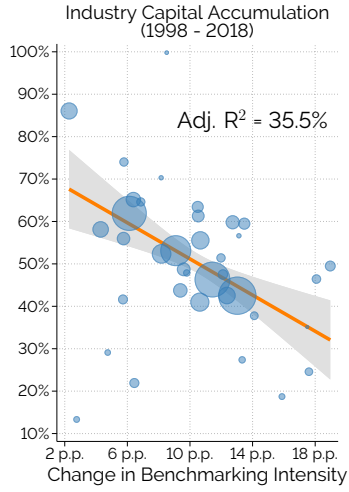
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▸ External validity MSCI ▸ NBER Industry results ▸ BEA Industry results ▸ China shock

Why are managers reducing investment? Incentives matter...

- Many firms use **Capital Efficiency Targets** in executives compensation
- 41% of Russell 3000 use these targets in **long-term incentive plans** (Reda & Tonello, 2015)

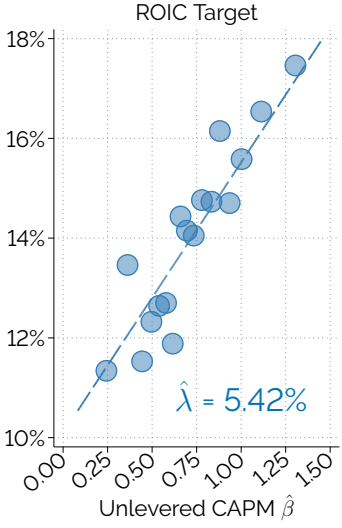
Why are managers reducing investment? Incentives matter...

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Our executive compensation program emphasizes performance-based compensation and is designed to tie directly to the drivers of value creation for the Company's shareholders, as summarized below.

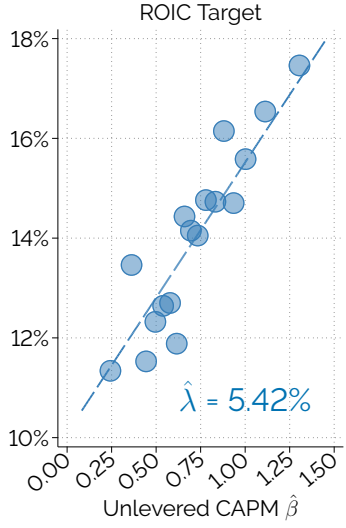
Key Elements		
Pay Element	**Form**	**Performance Metric**
Base Salary	Cash	> Base salary is set at market-competitive levels
Annual Cash Incentives	Cash	> 70% Absolute EPS(1) > 30% Strategic Goals
LTIP	50% - PSUs	> 60% three-year Relative EPS Growth(1) > 40% three-year Average Return on Invested Capital ("ROIC")(1) minus the Weighted Average Cost of Capital ("WACC")(1)
	50% - RSUs	> Stock price performance

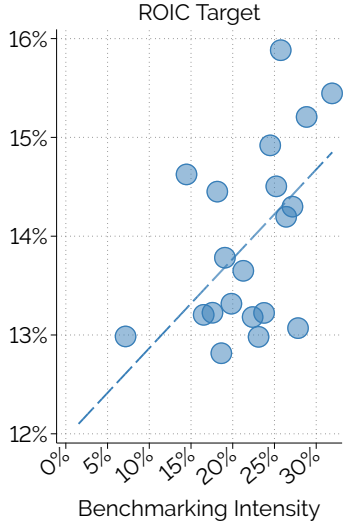
Source: Arrow Electronics' 2024 proxy statement (p. 44)



$$\text{ROIC Target} = \text{CAPM } \hat{\beta}^A + \text{Year} \times \text{Ind. FE} + \text{Firm FE} + \varepsilon$$

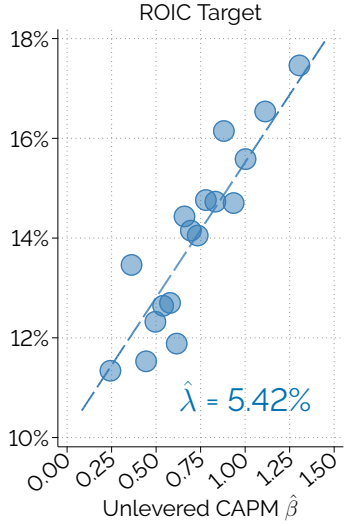
back | Details on ISS Incentive Lab data

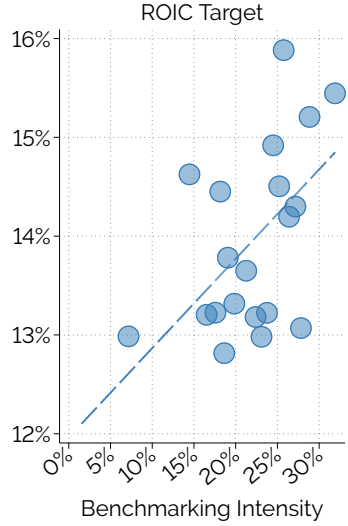


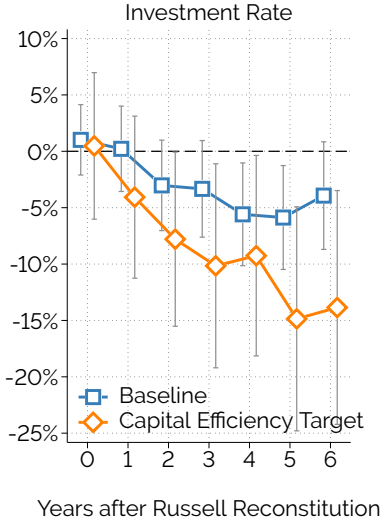


$$\text{ROIC Target} = \text{BMI} + \text{Year} \times \text{Ind. FE} + \text{Firm FE} + \varepsilon$$

back | Details on ISS Incentive Lab data







$$\Delta \log Inv_{i,t} = \Delta \text{BMI} + \Delta \text{BMI} \times 1(\text{Capital Efficiency Target}) + \text{Year} \times \text{Ind. FE} + \text{Cntrls} + \varepsilon$$

◂ back ◂ Details on ISS Incentive Lab data

Alternative Explanations?

Alternative mechanisms for declining investment?

- Fundamental risk exposure? ✗
  - Firm-level risk measures (Hassan et al., 2019)
  - Firm-level uncertainty shocks (Alfaro, Bloom, & Lin, 2024)

- Corporate Governance?
  - Governance scores
  - Common ownership (Rotemberg, 1984, Azar, 2012)
  - Activist campaigns (Appel, Gormley, & Keim, 2019)

- Price informativeness?
  - Sensitivity to Tobin's q (Antoniou et al., 2023)
  - Price Nonsynchronicity (Morck, Yeung, & Yu, 2000)
  - Bid-Ask Spread (Glosten & Milgrom, 1985)

▸ Risk exposure ▸ Fundamentals ▸ Common Ownership ▸ Activists ▸ Governance Scores ▸ Tobin's Q sensitivity ▸ Private info

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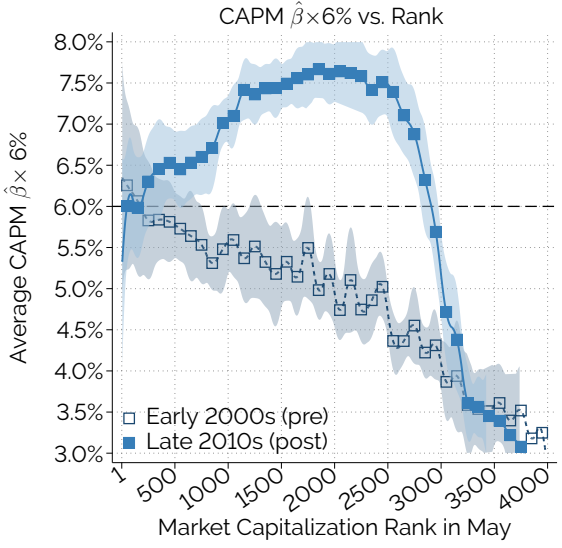
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  - Common ownership (Rotemberg, 1984, Azar, 2012)
  - Activist campaigns (Appel, Gormley, & Keim, 2019)

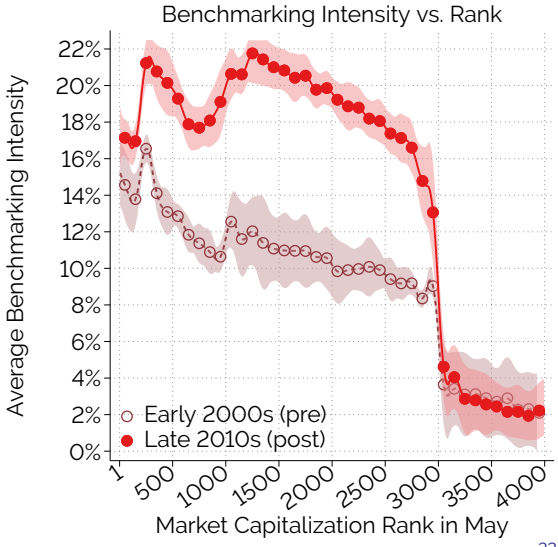
- Price informativeness? ✗/✓
  - Sensitivity to Tobin's q (Antoniou et al., 2023)
  - Price Nonsynchronicity (Morck, Yeung, & Yu, 2000)
  - Bid-Ask Spread (Glosten & Milgrom, 1985)

▸ Risk exposure ▸ Fundamentals ▸ Common Ownership ▸ Activists ▸ Governance Scores ▸ Tobin's Q sensitivity ▸ Private info

Effects of Benchmark-linked Ownership on Aggregate Investment

How can changes in XS of CAPM $\hat{\beta}$ affect aggregate investment if $\sum \omega \Delta\beta = 0$?





How can changes in XS of CAPM $\hat{\beta}$ affect aggregate investment if $\sum \omega \Delta\beta = 0$?

$$\Delta I_t / I_{t-1} = \sum_i \Delta I_{i,t} / I_{t-1} \approx \sum_i \pi_i \epsilon_i \underbrace{\mu_i \Delta\beta_{i,t} \lambda}_{\Delta \text{ cost of capital}}$$

- Intuition that changes cancel rests on two assumptions

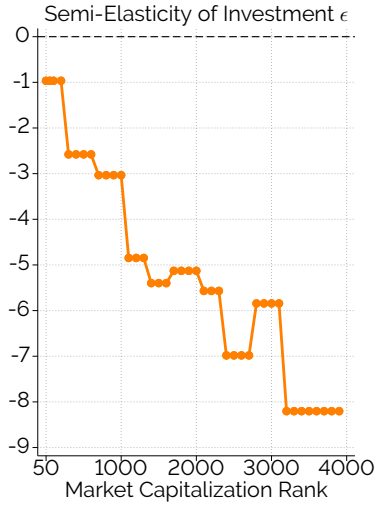
How can changes in XS of CAPM $\hat{\beta}$ affect aggregate investment if $\sum \omega \Delta\beta = 0$?

$$\Delta I_t / I_{t-1} = \sum_i \Delta I_{i,t} / I_{t-1} \approx \sum_i \pi_i \; \epsilon_i \; \underbrace{\mu_i \, \Delta\beta_{i,t} \, \lambda}_{\Delta \text{ cost of capital}}$$

- Intuition that changes cancel rests on two assumptions

A1 uniform **elasticity of investment** ϵ to cost of capital

Literature small firms are much more elastic than large firms
(Gertler & Gilchrist, 1994, Zwick & Mahon, 2017, Best et al., 2024)



▸ Measurement of Inv. Elasticities

How can changes in XS of CAPM $\hat{\beta}$ affect aggregate investment if $\sum \omega \Delta\beta = 0$?

$$\Delta I_t / I_{t-1} = \sum_i \Delta I_{i,t}/I_{t-1} \approx \sum_i \pi_i \epsilon_i \underbrace{\mu_i \Delta\beta_{i,t} \lambda}_{\Delta \text{ cost of capital}}$$

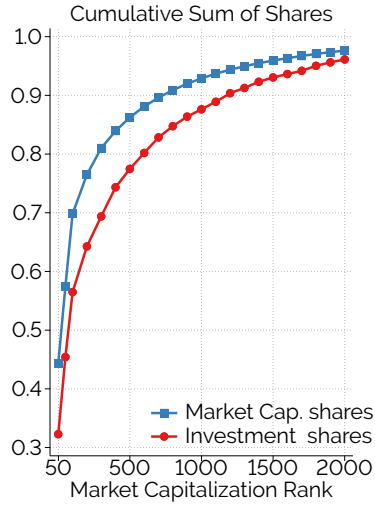
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A2 market cap shares ω coincide with **investment shares** π

Financials 18% market cap, 4% of CAPX



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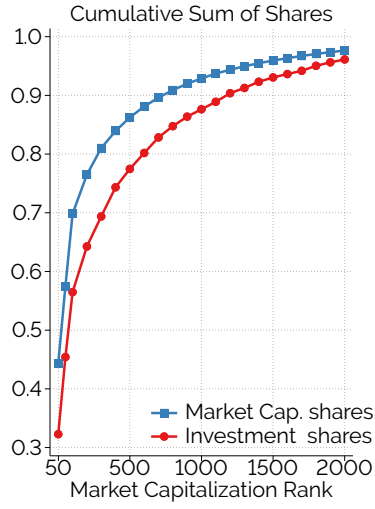
A1 uniform **elasticity of investment** ϵ to cost of capital

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A2 market cap shares ω coincide with **investment shares** π

Financials 18% market cap, 4% of CAPX

➠ **Changes in XS of CAPM $\hat{\beta}$ do not cancel out, but affect aggregate investment**



▸ Measurement of Inv. Elasticities

We use two complementary ways to aggregate across firms

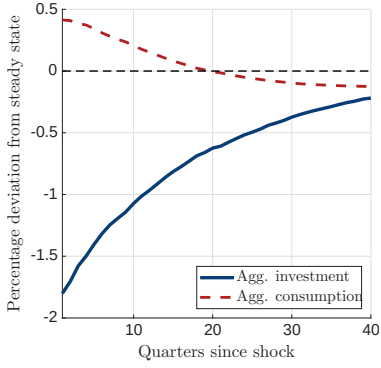
1. "Back-of-the-envelope" summation: agg. inv. $\downarrow$ 1.28%
   $\Delta I_t / I_{t-1} \approx \lambda \sum_i \pi_{i,t-1} \epsilon_{i,t} \mu_{i,t} \Delta\beta_{i,t}$
   - Avg. firm's response: 30%
   - Covariance$(\epsilon, \Delta\hat{\beta})$: 70%

▸ Details on Summation ▸ Estimation Table

We use two complementary ways to aggregate across firms

1. "Back-of-the-envelope" summation: agg. inv. ↓ 1.28%
   $\Delta I_t / I_{t-1} \approx \lambda \sum_i \pi_{i,t-1} \epsilon_{i,t} \mu_{i,t} \Delta\beta_{i,t}$
   - Avg. firm's response: 30%
   - Covariance$(\epsilon, \Delta\hat{\beta})$: 70%

2. Calibrate heterogeneous firm GE model: agg. inv. ↓ 1.85%
   - Convex & fixed adjustment costs
   - Shock to perceived cost of capital



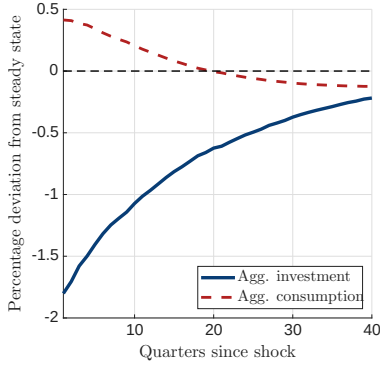
▸ Details on Summation ▸ Estimation Table ▸ General equilibrium model
▸ Counterfactual WACC

We use two complementary ways to aggregate across firms

1. "Back-of-the-envelope" summation: agg. inv. ↓ 1.28%
   $\Delta I_t / I_{t-1} \approx \lambda \sum_i \pi_{i,t-1} \epsilon_{i,t} \mu_{i,t} \Delta\beta_{i,t}$
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   - Convex & fixed adjustment costs
   - Shock to perceived cost of capital

- Caveats: competition, innovation, etc.
  - Firm entry and exit decisions
  - Disappearing public firms
  - Misallocation



▸ Details on Summation ▸ Estimation Table ▸ General equilibrium model ▸ Counterfactual WACC

Conclusion

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- **Benchmark-linked ownership** raises CAPM $\hat{\beta}$ and the perceived cost of equity
- Managers respond to **increase in perceived cost of equity** by **reducing investment**

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- **Benchmark-linked ownership** raises CAPM $\hat{\beta}$ and the perceived cost of equity
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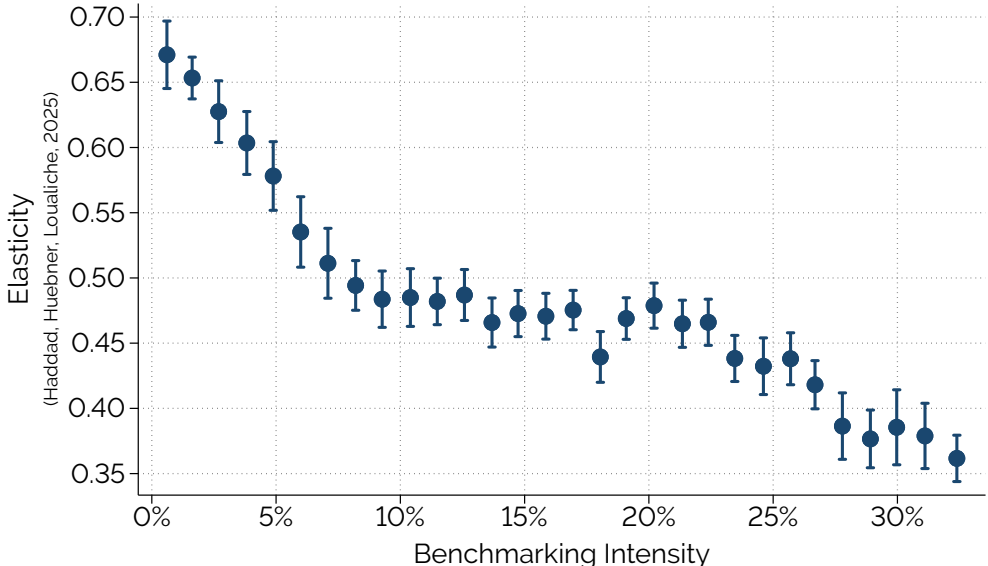
> *"If you use some of the tools I learned in my **MBA class**, like the **capital asset pricing model**, they did teach that **back in the 80s** by the way, so it's been around for a while. I think our cost of equity is around 10%."*
>
> — CFO, Qorvo (Q4 2015)

➠ Maybe it's time that we update our curricula...

Appendix

BMI vs. stock-level elasticity estimates of Haddad, Huebner, Loualiche ('25)

back to Data | back to MBA | back to Fax



Firm managers use CAPM to estimate cost of equity in practice

back to MBA

Surveys

- Graham & Harvey (2001): 74% of CFOs (almost) always use CAPM
- Jacobs & Shivdasan (2012): 90% use CAPM in DCF
- Mukhlynina & Nyborg (2016): 76% (almost) always use CAPM
- Graham (2022): 80% of CFOs (almost) always use CAPM

Revealed Preferences

- Krüger et al. (2015): managers use CAPM in capital budgeting
- Jaganathan et al. (2016): CAPM βs explain discount rates of CFOs
- Dessaint et al. (2020): managers use CAPM to value acquisitions
- Gormsen & Huber (2024): CAPM explains 50% of variation in perc. cost of capital
- Cho & Salarkia (2025): managers use CAPM to time repurchases

Other Professions

- Berk & Binsbergen (2015); Barber et al. (2016): mutual fund investors use CAPM
- Jensen (2023); Decaire & Graham (2024): stock analysts use CAPM
- Kontz (2025): utility regulators use CAPM

*"On the equity component, we calculate cost of equity using the **capital asset pricing model** using historical data and market risk premiums. Our current cost of equity is running around 9%."*

— CFO, Archer Daniels Midland (Q1 2012)

Executives Quotes on usage of CAPM

Executives Commenting on Index Inclusion

Appendix 2

Summary Statistics Beta Panel

back

	count	mean	sd	min	p25	p50	p75	max
Benchmarking Intensity (BMI, %)	929,105	13.27	9.25	0.00	3.67	14.20	21.14	99.07
Institutional Ownership Ratio (IOR, quarterly %)	311,355	53.99	31.29	0.00	27.06	57.02	80.65	130.50
Avg. Implied Cost of Capital (ICC, %)	696,830	9.37	2.98	2.16	7.57	9.06	10.85	20.42
CAPM $\hat{\beta}$ (Welch)	886,432	0.95	0.47	-0.07	0.60	0.94	1.27	2.65
Unlevered CAPM $\hat{\beta}$ (Welch)	886,432	0.58	0.43	-1.95	0.22	0.52	0.86	3.30
CAPM $\hat{\beta}$ (21-day)	925,786	0.98	1.04	-4.93	0.36	0.91	1.50	13.63
CAPM $\hat{\beta}$ (252-day)	910,903	0.97	0.59	-0.60	0.54	0.94	1.33	3.69
CAPM $\hat{\beta}$ (156-week)	882,232	1.04	0.61	-0.37	0.60	0.98	1.39	4.07
CAPM $\hat{\beta}$ (36-month)	830,936	1.19	0.91	-1.47	0.57	1.05	1.64	5.92
Cash-Flow Beta (ROE, quarterly)	248,879	0.46	1.42	-8.25	-0.15	0.30	0.92	9.16
Consumption Beta (KKL, annual)	46,930	1.10	0.94	-2.62	0.51	1.01	1.66	5.46
Log Volume	929,038	17.64	2.59	10.67	15.74	17.75	19.57	23.65
Equity Ratio (%)	929,105	59.32	28.28	1.42	37.22	64.00	84.13	99.41
Amihud Illiquidity	920,947	1.44	9.30	0.00	0.00	0.01	0.12	216.27
Bid-Ask Spread	899,882	0.01	0.01	0.00	0.01	0.01	0.02	0.13
Log Market Equity	929,073	6.28	1.86	1.53	4.92	6.12	7.49	11.93
Momentum Return (12-month, %)	889,951	14.04	64.31	-96.90	-19.17	6.22	32.69	1478.13

Summary Statistics Investment Panel

back

	count	mean	sd	min	p25	p50	p75	max
CAPX/PPEGT (%)	37,787	14.90	15.09	0.58	5.92	9.98	17.71	82.54
Benchmarking Intensity (BMI, %)	38,628	15.04	9.42	0.00	7.58	17.02	22.63	65.98
Institutional Ownership Ratio (IOR, %)	38,511	64.92	27.58	0.01	45.94	70.34	86.67	118.00
Tobin's q^{tot}	38,616	1.37	1.74	-0.55	0.43	0.84	1.59	14.91
Leverage (%)	37,947	54.78	29.77	5.82	33.10	51.92	70.37	169.38
Current Ratio (%)	38,619	284.73	231.02	52.10	145.69	212.50	330.50	1622.90
Log(Market Cap.)	38,581	6.80	1.71	2.54	5.54	6.64	7.91	11.23
Cash Flow (%)	37,904	-1.48	257.30	-7146.02	6.68	18.77	42.01	295.32
Firm Age (years)	38,628	23.67	18.50	1.42	10.42	17.42	32.42	92.42

Value-weighted CAPM $\beta \equiv \sum_i \omega_i \hat{\beta}_i = \bar{\beta} + \mathbf{cov}\left(\omega_i, \hat{\beta}_i\right) = 1$

back

Value-weighted CAPM $\beta \equiv \sum_i \omega_i \hat{\beta}_i = \bar{\beta} + \mathbf{cov}\left(\omega_i, \hat{\beta}_i\right) = 1$

back

	Small-Cap	Micro-Cap
$\mathbb{1}\{\text{Year} > 2002\}$	**0.43**	**0.40**
	(0.05)	(0.05)
Constant	0.81	0.56
	(0.04)	(0.03)
Avg. Market Cap. Share	0.06	0.02
Avg. CAPX Share	0.09	0.04
# Stocks	**950**	**1768**
Observations	360	360

$$\bar{\beta}_t^{Group} = \alpha + \mathbb{1}\{\text{Year} > 2002\}_t + \varepsilon_t$$

Value-weighted CAPM $\beta \equiv \sum_i \omega_i \hat{\beta}_i = \bar{\beta} + \mathbf{cov}\left(\omega_i, \hat{\beta}_i\right) = 1$

back

	Large-Cap	Small-Cap	Micro-Cap
$\mathbb{1}\{\text{Year} > 2002\}$	**0.19**	0.43	0.40
	(0.03)	(0.05)	(0.05)
Constant	0.92	0.81	0.56
	(0.03)	(0.04)	(0.03)
Avg. Market Cap. Share	0.16	0.06	0.02
Avg. CAPX Share	0.20	0.09	0.04
# Stocks	**589**	950	1768
Observations	360	360	360

$$\bar{\beta}_t^{Group} = \alpha + \mathbb{1}\{\text{Year} > 2002\}_t + \varepsilon_t$$

Value-weighted CAPM $\beta \equiv \sum_i \omega_i \hat{\beta}_i = \bar{\beta} + \mathbf{cov}\left(\omega_i, \hat{\beta}_i\right) = 1$

back

	Mega-Cap	Large-Cap	Small-Cap	Micro-Cap
$\mathbb{1}\{\text{Year} > 2002\}$	**0.04**	0.19	0.43	0.40
	(0.02)	(0.03)	(0.05)	(0.05)
Constant	1.00	0.92	0.81	0.56
	(0.02)	(0.03)	(0.04)	(0.03)
Avg. Market Cap. Share	0.34	0.16	0.06	0.02
Avg. CAPX Share	0.40	0.20	0.09	0.04
# Stocks	**269**	589	950	1768
Observations	360	360	360	360

$$\bar{\beta}_t^{Group} = \alpha + \mathbb{1}\{\text{Year} > 2002\}_t + \varepsilon_t$$

Value-weighted CAPM $\beta \equiv \sum_i \omega_i \hat{\beta}_i = \bar{\beta} + \mathbf{cov}\left(\omega_i, \hat{\beta}_i\right) = 1$

back

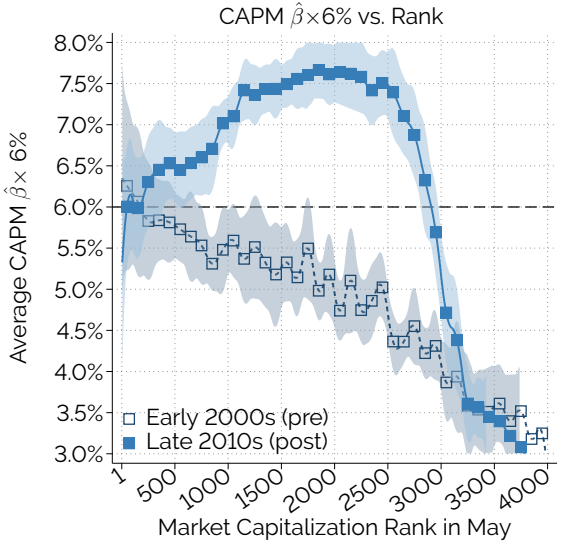
	Top-50	Mega-Cap	Large-Cap	Small-Cap	Micro-Cap
$\mathbb{1}\{\text{Year} > 2002\}$	**-0.13**	0.04	0.19	0.43	0.40
	(0.03)	(0.02)	(0.03)	(0.05)	(0.05)
Constant	1.07	1.00	0.92	0.81	0.56
	(0.03)	(0.02)	(0.03)	(0.04)	(0.03)
Avg. Market Cap. Share	**0.42**	0.34	0.16	0.06	0.02
Avg. CAPX Share	0.27	0.40	0.20	0.09	0.04
# Stocks	**50**	269	589	950	1768
Observations	360	360	360	360	360

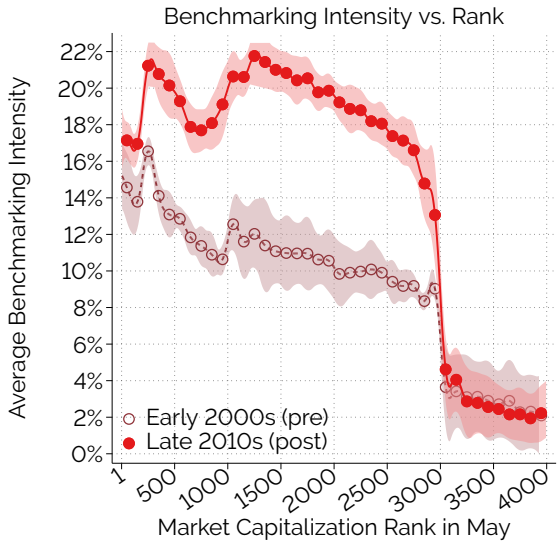
$$\bar{\beta}_t^{Group} = \alpha + \mathbb{1}\{\text{Year} > 2002\}_t + \varepsilon_t$$

Time series of equal-weighted average β

Fact 1: Benchmarking intensity positively correlates with CAPM $\hat{\beta}$s

◂ Fax ◂ Agg.

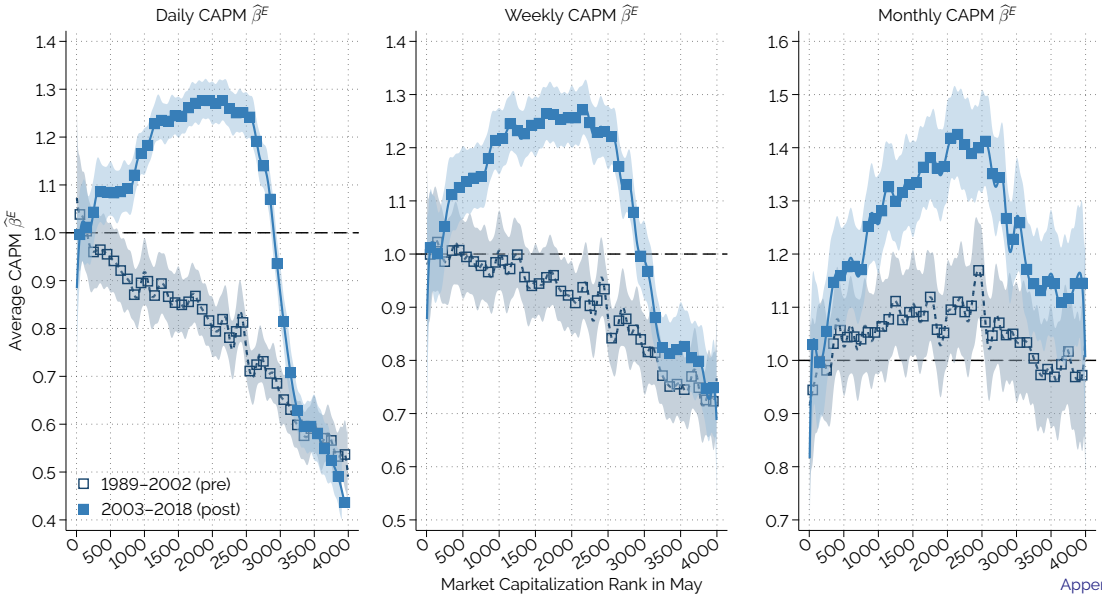




▸ Value-weighted β is 1 ▸ Top-50 $\hat{\beta}$ ▸ Scholes-Williams β ▸ Other β freq. ▸ Instit. Ownership ▸ Cash flow βs ▸ Asset βs

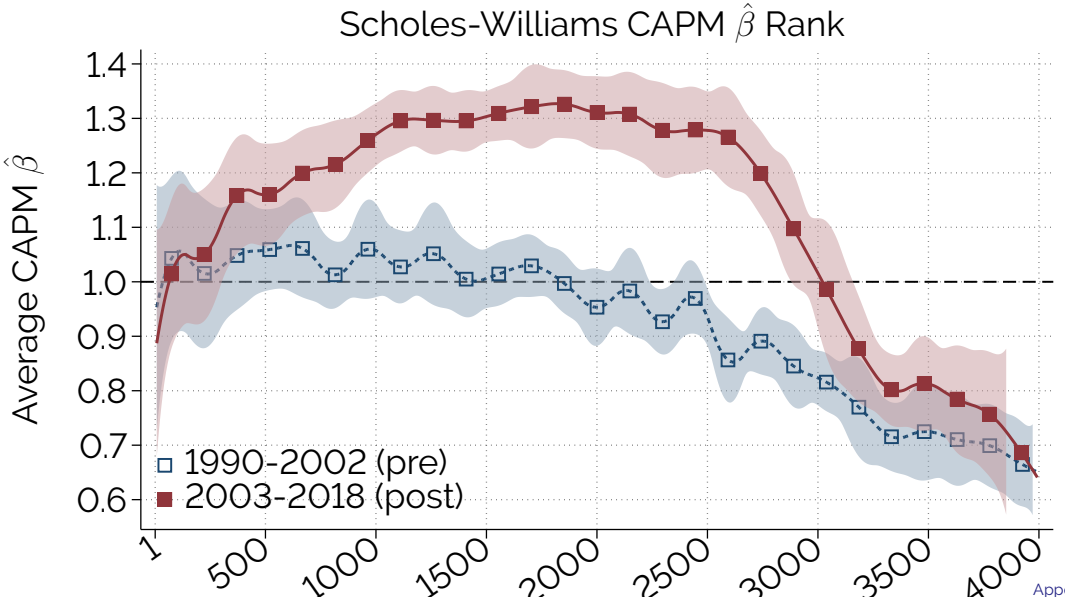
Appendix 6

Fact 1 is robust across CAPM $\hat{\beta}$ estimation frequencies



Appendix 7

Fact 1 is robust to adjustment for infrequent trading



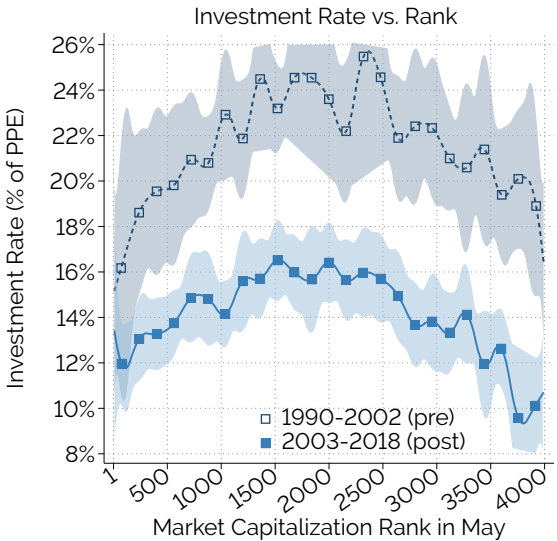
Appendix 8

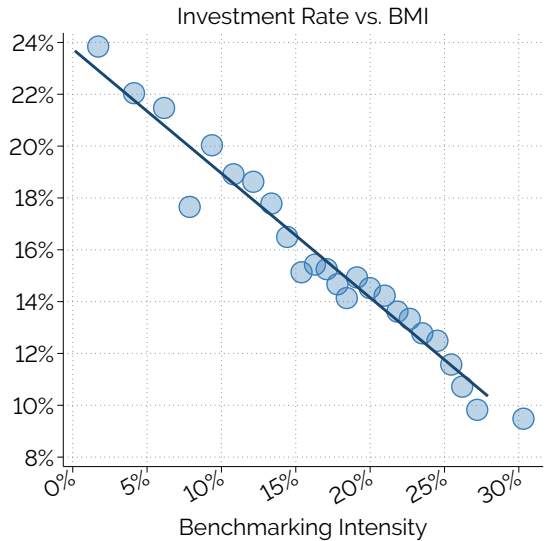
Effect of BMI on $\hat{\beta}$ declines with frequency, consistent with temporary price pressure

◂ Facts ◂ β Sum.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Dep. Var.: CAPM $\hat{\beta}$							
	21-Daily Returns		252-Daily Returns		156-Weekly Returns		36-Monthly Returns	
BMI (in %)	0.032***	0.032***	0.029***	0.029***	0.019***	0.018***	0.011***	0.011***
	(0.001)	(0.001)	(0.002)	(0.002)	(0.001)	(0.001)	(0.002)	(0.002)
Time FE	✓		✓		✓		✓	
Time × FF49 × Size FE		✓		✓		✓		✓
Adj. R^2	0.09	0.17	0.23	0.46	0.10	0.37	0.03	0.26
Observations	939,104	927,544	81,784	80,693	79,109	78,083	74,485	73,566

Fact 2: Benchmarking intensity negatively correlates with investment rate



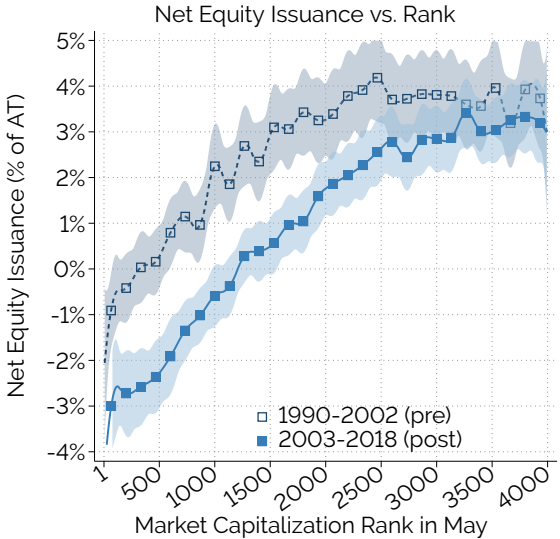


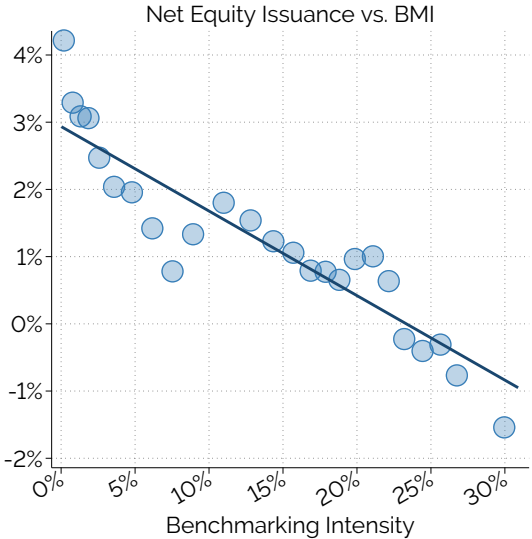
$$\text{CAPX/PPE} = \textbf{BMI} + \text{Time} \times \text{Size FE} + \varepsilon$$

▸ Other investment rate definitions

Appendix 10

Fact 3: Benchmarking intensity negatively correlates with equity issuance

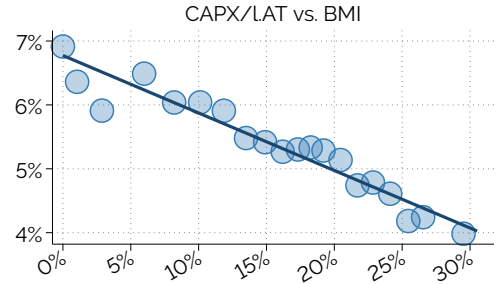


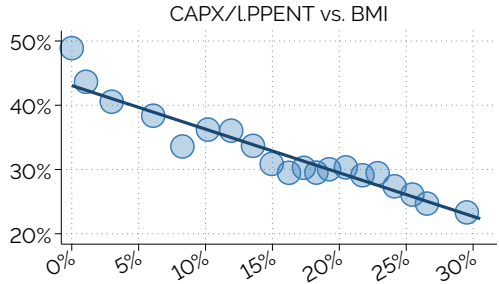


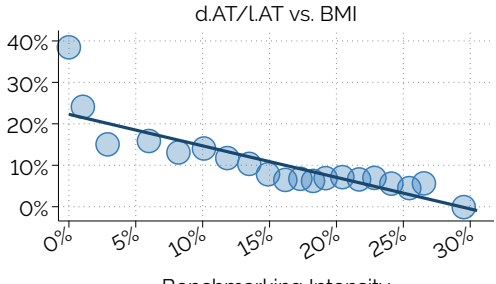
$$\text{Net Issu.} = \textbf{BMI} + \text{Time} \times \text{Size FE} + \varepsilon$$

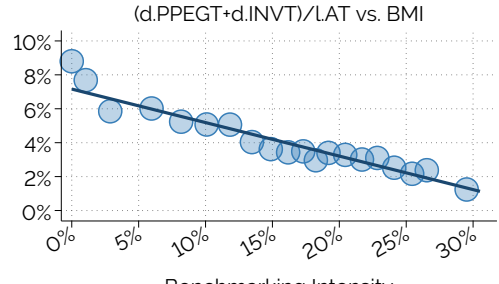
Appendix 11

Fact 2 is robust to different investment rate definitions



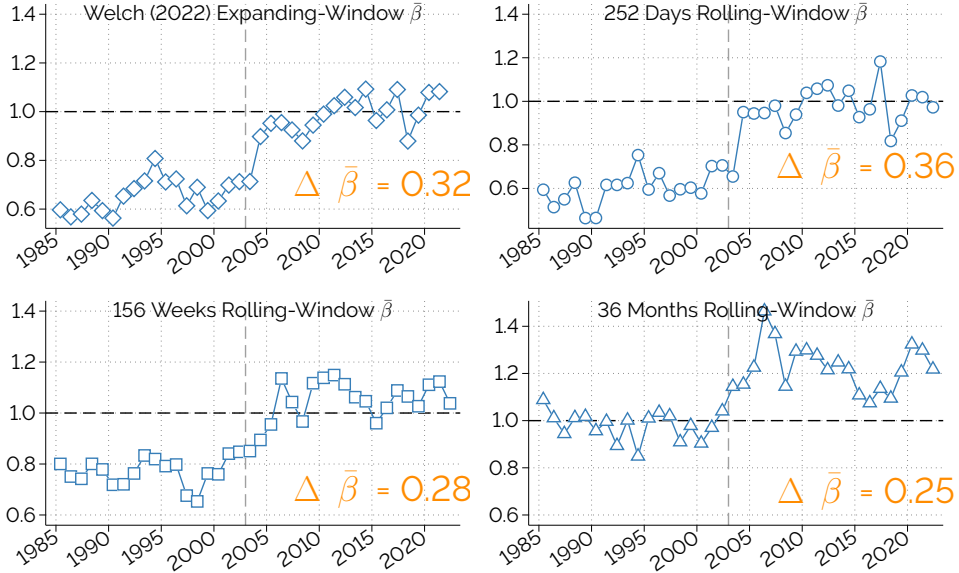






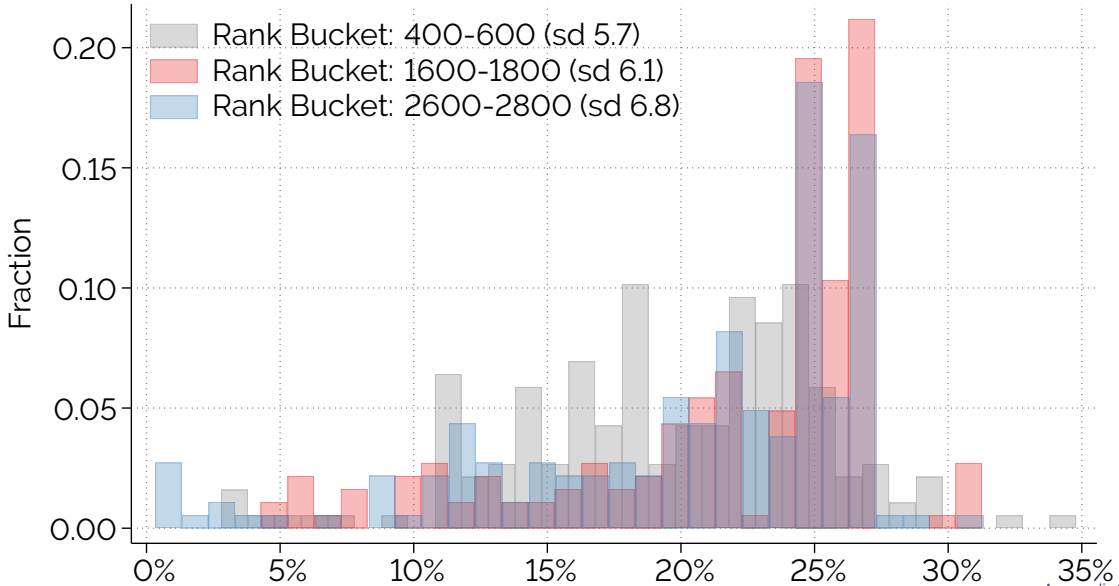
Equal-weighted average CAPM $\bar{\beta}$ increased for all frequencies

◂ Fax 1 ◂ $\hat{\beta}$ summary



Variation within Time × Size Cell in June 2010

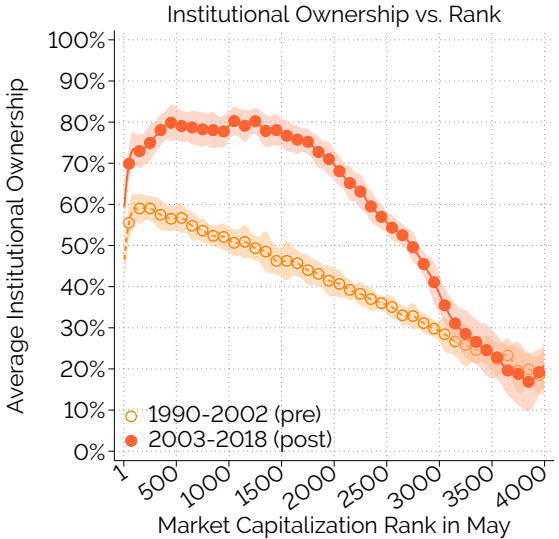
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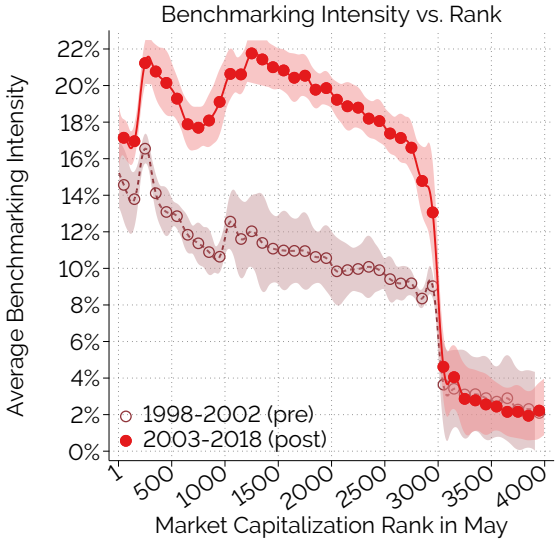


Institutional ownership ratio does not show similar kinks

Facts

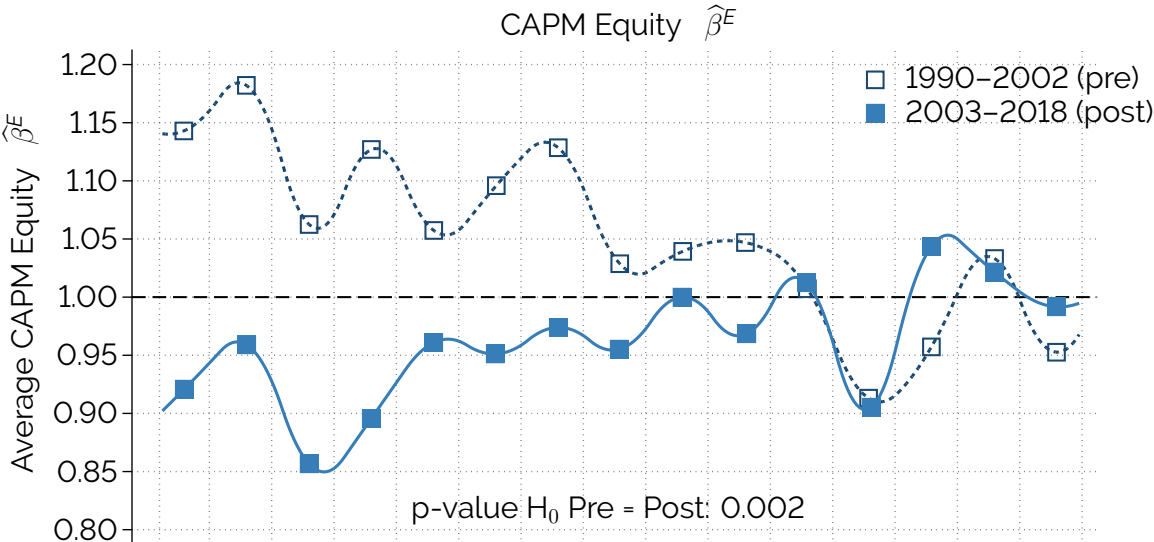
Beta DiD





CAPM $\hat{\beta}$s of 50 largest firms by market cap declined

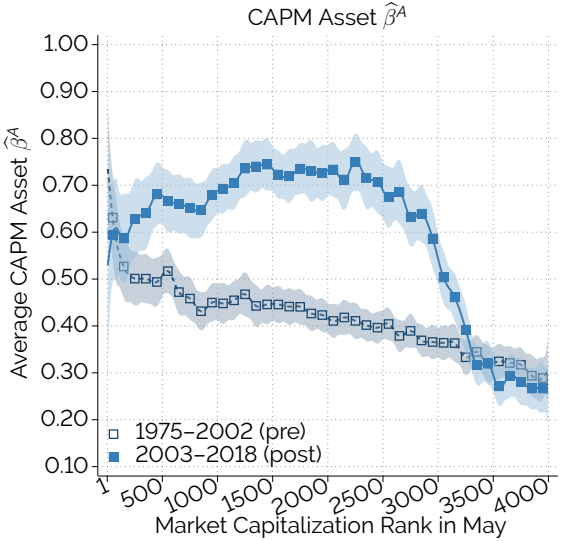
back

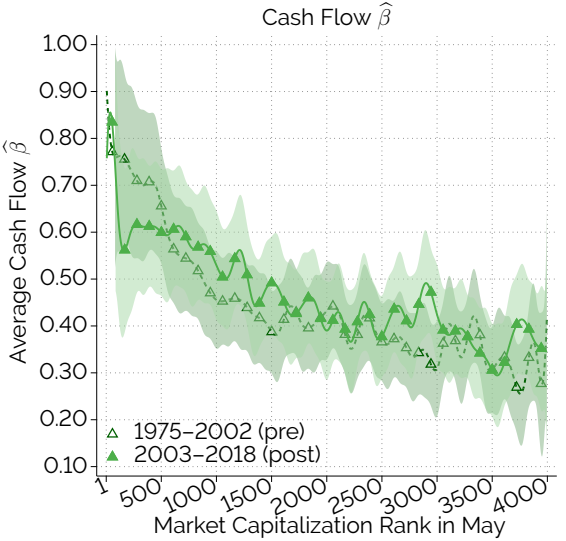


Appendix 16

Leverage cannot explain pattern, cash flow βs did not change

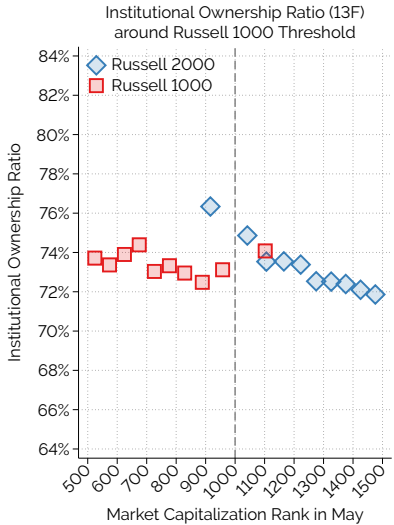
back to facts | back to beta summary

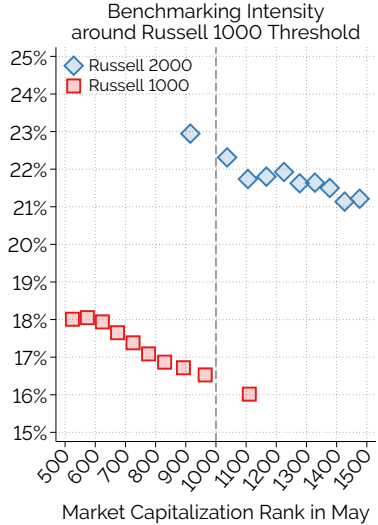




Russel Reconstitution: No change in level of Institutional Ownership but change from Active to Passive Ownership

◂ Russell ◂ Beta DiD





back

The CAPM β of a stock is equal to

$$\beta_{i0}^{CAPM} = \frac{Cov_0\left(\phi_i^z \Delta z_1 + \sum_k [\phi_{k,S(i)}^\lambda + \phi_{B_k(i)}^\lambda]\Delta\lambda_1^{B_k} + \nu_{i1}, \sum_j \phi_j^z \Delta z_1 + \sum_k [\phi_{k,S(j)}^\lambda + \phi_{B_k(j)}^\lambda]\Delta\lambda_1^{B_k} + \nu_{j1}\right)}{Var_0\left(\sum_j \phi_j^z \Delta z_1 + \sum_k [\phi_{k,S(j)}^\lambda + \phi_{B_k(j)}^\lambda]\Delta\lambda_1^{B_k} + \nu_{j1}\right)}$$

$$= \frac{\phi_i^z(\mathbf{1}'\phi^z)\sigma_z^2 + [\phi_{S(i)}^\lambda + \phi_{B(i)}^\lambda]'\Omega_\lambda[\phi_S^\lambda + \phi_B^\lambda]\sigma_\lambda^2 + \sigma_{\nu,i}^2}{(\mathbf{1}'\phi^z)^2\sigma_z^2 + [\phi_S^\lambda + \phi_B^\lambda]'\Omega_\lambda[\phi_S^\lambda + \phi_B^\lambda]\sigma_\lambda^2 + \sum_j \sigma_{\nu,j}^2}$$

where

$$\phi_i^z = b_i\left(1 - \gamma\Lambda(\lambda_0^{\mathcal{B}})\sigma_z^2\left(1 - \frac{c}{a+c}b'BMI_0\right)\right)$$

$$\phi_{k,S(i)}^\lambda = \gamma\Lambda_0\left\{\frac{c}{a+c}\left[b'\mathbf{1}_{B_k} + \Lambda_0\left(1 - \frac{c}{a+c}b'BMI_0\right)\right]b_i z_0\sigma_z^2 - \Lambda_0\left(1 - \frac{c}{a+c}\right)\left(b_i z_0\sigma_z^2 + \varepsilon_{i0}\sigma_\varepsilon^2\right)\right\}$$

$$\phi_{k,B(i)}^\lambda = \gamma\Lambda_0\left\{\frac{c}{a+c}\left[b'\mathbf{1}_{B_k}(i) + \Lambda_0\left(1 - \frac{c}{a+c}\right)BMI_{0i}\right]\right\}\varepsilon_{i0}\sigma_\varepsilon^2$$

Benchmark-linked investing raises pairwise corr. of benchmark stocks

back

- Theories predict that benchmark-linked investing causes "excess" co-movement
  - e.g., Basak & Pavlova ('13), Garcia & Bond ('19), Grégoire ('19), Chabakauri & Rytchkov ('21), Baruch & Zhang ('22), Buffa & Hodor ('23), Davies ('24), Kashyap et al (2021, Appendix D)

$$\text{CAPM } \beta_i = \frac{\text{Cov}\left[R_i, R_m\right]}{\text{Var}\left[R_m\right]} = \frac{\sum_j w_j \, \sigma_i \, \sigma_j \, \rho_{i,j}}{\text{Var}\left[R_m\right]}$$

Sim $w \sim \text{Lognormal}(0, 1)$, $\sigma_j = \sigma_i = 0.05$, $\bar{\rho}_{ij} = 0.1$

Benchmark-linked investing raises pairwise corr. of benchmark stocks

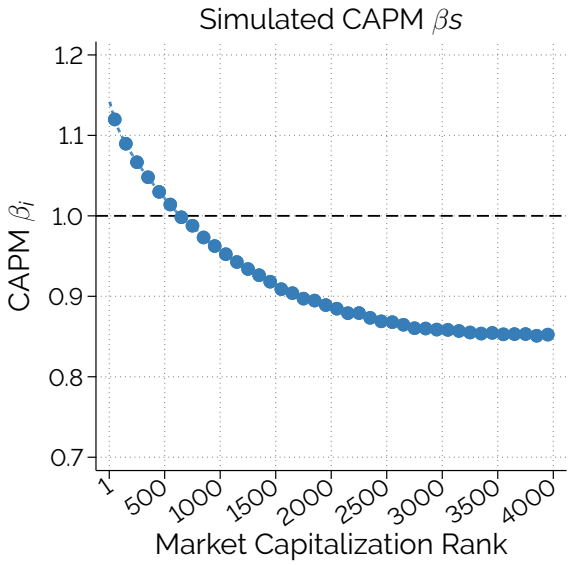
back

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Sim $\Delta\rho = 0.1$ among Russell 2000



Benchmark-linked investing raises pairwise corr. of benchmark stocks

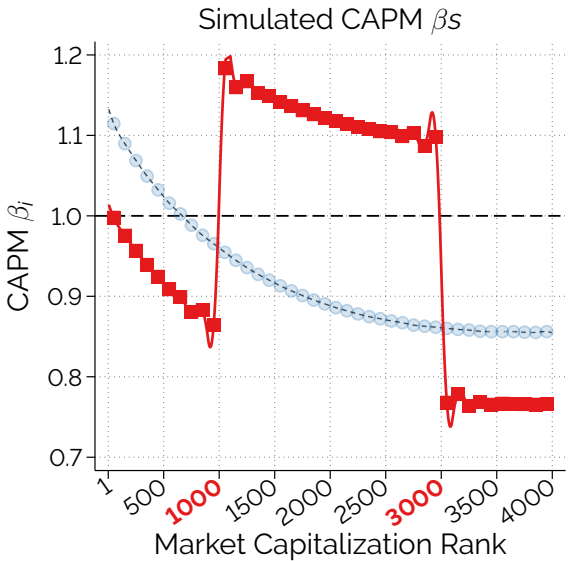
back

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Sim $w \sim \text{Lognormal}(0, 1)$, $\sigma_j = \sigma_i = 0.05$, $\bar{\rho}_{ij} = 0.1$

Sim $\Delta\rho = 0.1$ among Russell 2000



Changes in BMI strongly correlate with trading volume during reconstitution

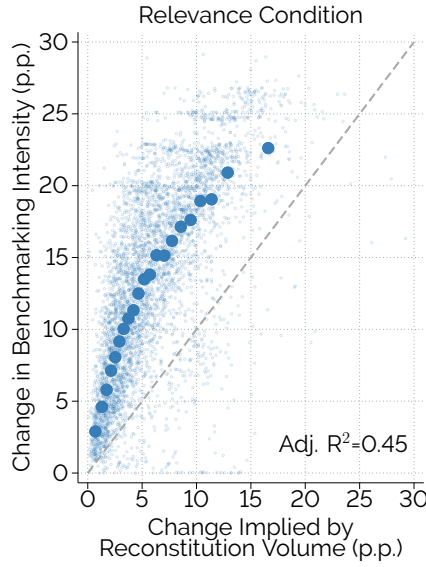
back

- Chinco & Sammon (2024) note that

$$\underbrace{\text{Recon. Volume} \times \text{Price}}_{\text{value of shares purchased}} = \underbrace{\text{Index Weight} \times \text{AUM benchmarked}}_{\text{value of required position}}$$

- Note further that

$$\begin{aligned}\text{BMI} &= \frac{\text{Index Weight} \times \text{AUM benchmarked}}{\text{Market Cap of Stock}} \\ &= \frac{\text{Recon. Volume} \times \text{Price}}{\text{Market Cap of Stock}}\end{aligned}$$



Rank proxy accurately predicts membership around 1000/2000 cutoff

back

	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent variable: Russell 2000 Membership in June					
	Window width: 300 ranks			Window width: 150 ranks		
	1998-2018	1998-2006	2007-2018	1998-2018	1998-2006	2007-2018
$\mathbb{1}\{\text{Rank} > \text{cutoff in May}\}$	0.843***	0.924***	0.771***	0.830***	0.867***	0.751***
	(0.056)	(0.016)	(0.036)	(0.058)	(0.031)	(0.035)
Year Fixed Effect	✓	✓	✓	✓	✓	✓
Baseline Controls	✓	✓	✓	✓	✓	✓
Banding Controls	✓		✓	✓		✓
Adj. R^2	0.91	0.95	0.94	0.88	0.92	0.92
Observations	15,032	4,456	10,130	8,721	2,183	6,268

$$\text{Russell 2000 Member in June}_{i,t} = \alpha_t + \mathbb{1}\{\text{Rank} > \text{cutoff in May}\}_{i,t} + \text{Controls}_{i,t} + \varepsilon_{i,t}$$

NB: Pavlova & Sikorskaya (2023) proprietary rank variable coefficient ≈ 0.87 in (5), ≈ 0.86 in (6)

Rank proxy allows us to replicate main result in Pavlova & Sikorskaya (2023)

◂ Russell identification ◂ Price effect on ICC

	Total stock return in period				
Period	April	May	**June**	July	April-July
Panel A: Original Estimates of Pavlova & Sikorskaya (2023) from their Table A.19					
ΔBMI	0.151*	0.192***	**0.271****	0.037	0.677***
	(0.075)	(0.055)	**(0.102)**	(0.116)	(0.184)
Observations	14,547	14,547	14,547	14,547	14,547
Panel B: Replication of Estimates Using Ben-David et al. (2019) Ranking Variable Proxy					
ΔBMI	0.241**	0.186***	**0.267*****	0.035	0.787***
	(0.117)	(0.068)	**(0.102)**	(0.140)	(0.260)
Observations	13,232	13,233	13,232	13,157	13,266
p-value (Panel B = Panel A)	0.45	0.94	**0.97**	0.99	0.68

Benchmarking intensity explains up to 26% of variation in $\hat{\beta}$

◂ β DiD ◂ Placebo

$$\text{CAPM } \hat{\beta}_{i,t+1} = \textbf{BMI}_{i,t} + \text{Year} \times \text{Ind FE} + \text{Stock FE} + \varepsilon_{i,t+1}$$

	Dependent variable: CAPM $\hat{\beta}$ (using 252 daily returns)					
BMI (in %)	0.027***		0.026***	0.025***	0.022***	**0.023*** **
	(0.002)		(0.002)	(0.001)	(0.002)	**(0.001)**
IOR (in %)		**0.006*** **	**0.000**			
		(0.001)	**(0.000)**			
Constant	0.593***	0.588***				
	(0.022)	(0.021)				
Adj. R^2	**0.26**	**0.13**	**0.26**	0.49	0.55	0.74
Within R^2				**0.29**	**0.18**	**0.17**
FF49 × Yr FE				✓		
FF49 × Size × Yr FE					✓	✓
Stock FE						✓
Observations	77,445	77,443	77,445	76,823	76,277	74,922

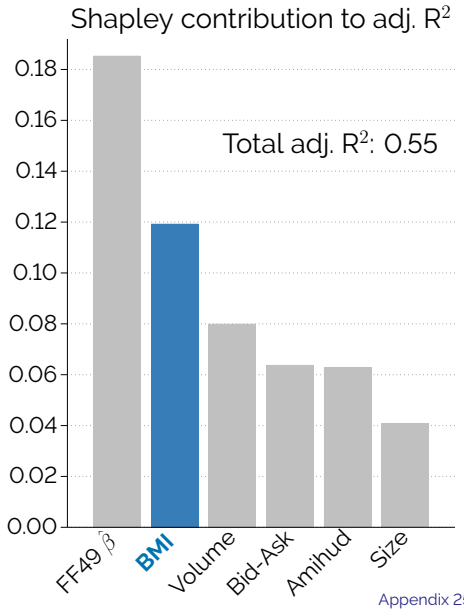
Appendix 24

BMI explains 12% of unique variation in CAPM $\hat{\beta}$s

back

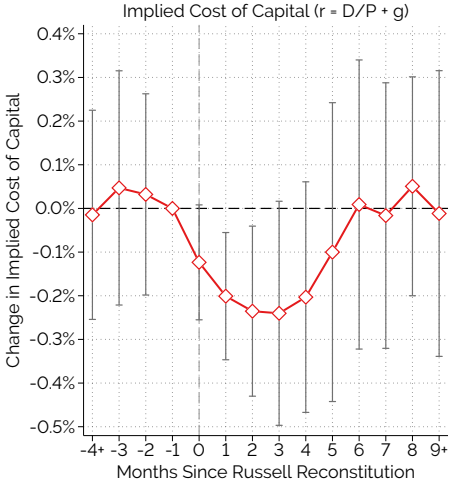
$$\text{CAPM } \hat{\beta} = \text{FF49 Ind. } \hat{\beta} + \textbf{BMI} + \text{Trading Volume} + \text{Amihud Illiq.} + \text{Size} + \varepsilon$$

NB "Bad Control" problem: trading volume and liquidity are partly determined by **benchmarking intensity**



Effect on implied cost of capital is small and reverts within 6 months

back



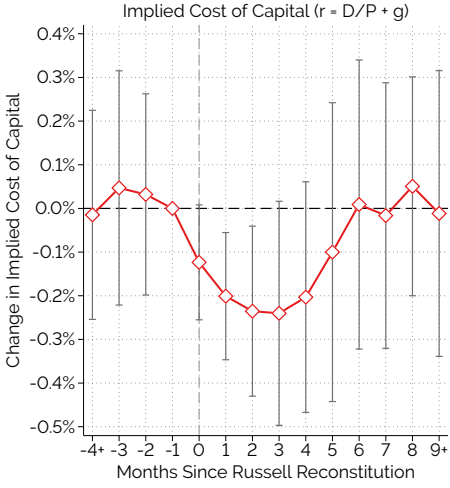
$$\frac{P^{Post}}{P^{Pre}} - 1 = \frac{-\Delta r}{r^{Pre} + \Delta r - g} = \frac{0.21\%}{8.9\% - 0.21\% - 1.6\%} = \underset{(1.17)}{3.00}^{**}\%$$

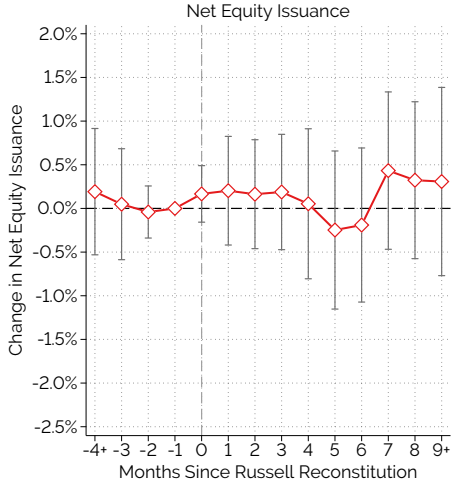
- CAPM SML in ICC
- Exec. comments on benchmark inclusion
- Replication Pavlova & Sikorskaya ('23)

Appendix 26

Effect on implied cost of capital is small and reverts within 6 months

back





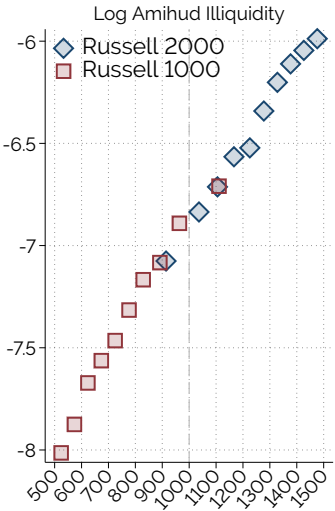
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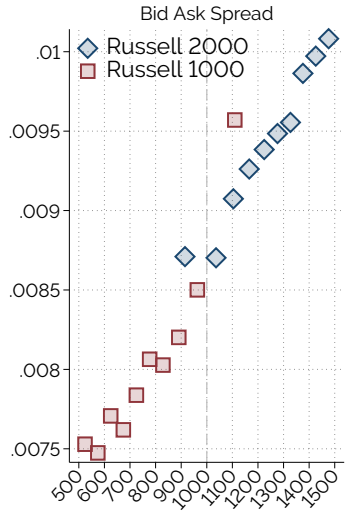
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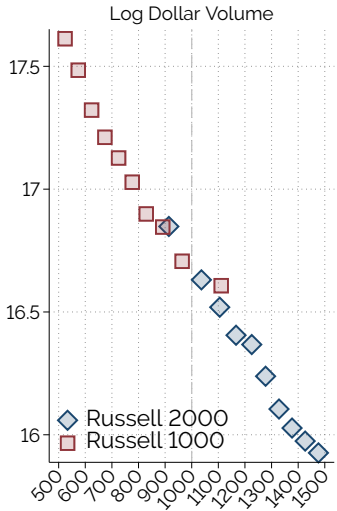
Appendix 26

No difference in liquidity around Russell 1000 cutoff

back

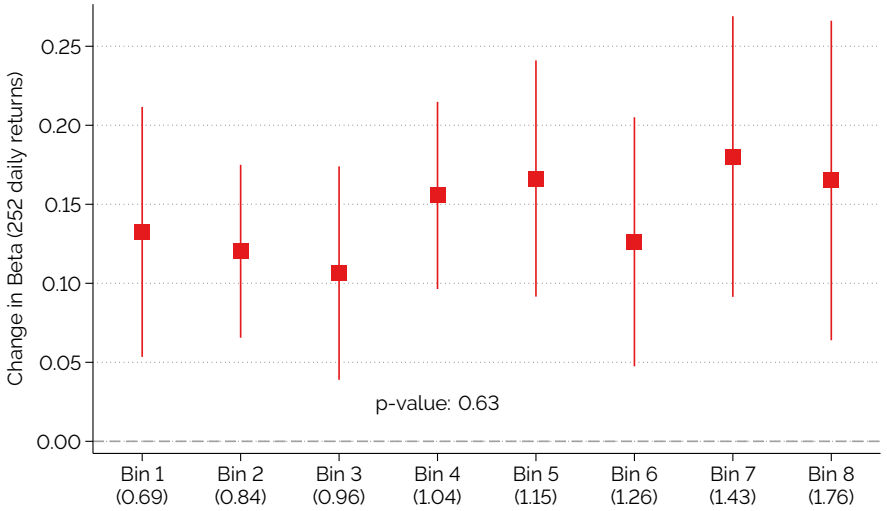






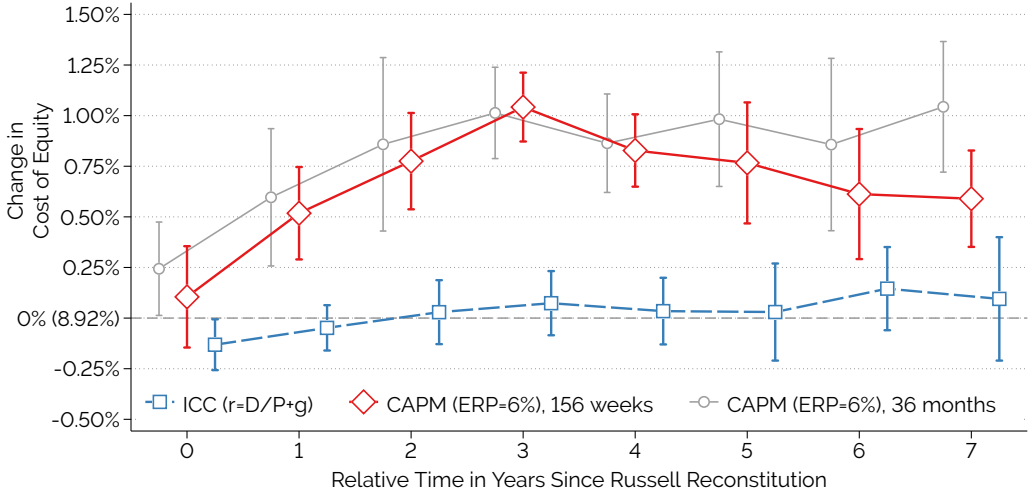
No difference in treatment effect based on pre-treatment level of $\hat{\beta}$

back



Effect of exog. BMI increase on CAPM $\hat{\beta}$ is persistent

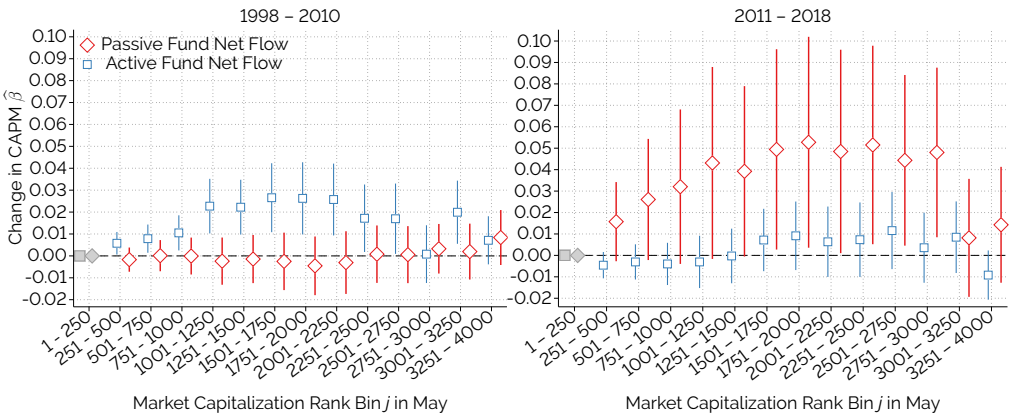
Transition Persistence · Beta · Inv.



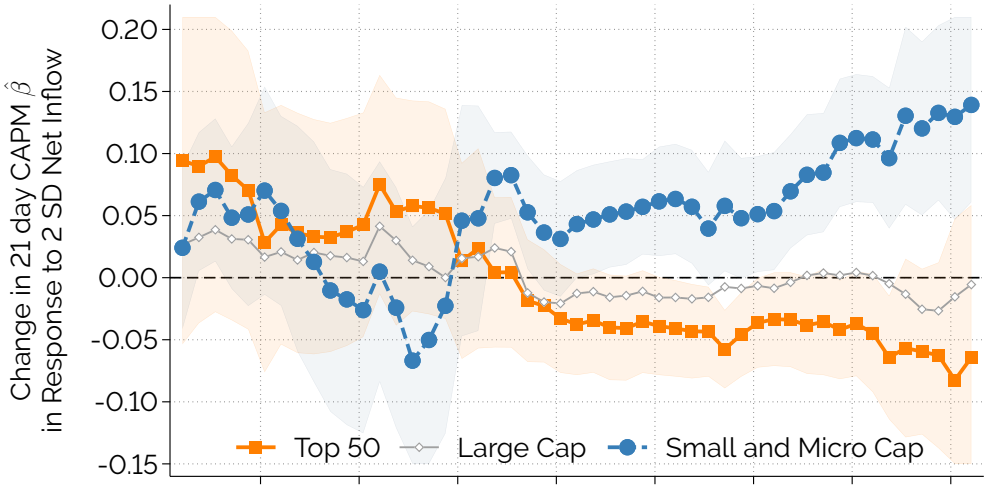
$$\text{CAPM } \hat{\beta}_{i,t+h} \times 6\% = \gamma^h_{t,s} + \gamma^h_1 \Delta \text{BMI}_{i,t} + X'_{i,t}\zeta^h + \epsilon_{i,t+h}, \quad h = 0, \ldots, 7$$

$$\Delta\text{CAPM}\ \hat{\beta}_{i,t} = \alpha_i + \alpha_t + \sum_{j \notin \{250\}} \gamma_j^{Active}\, \mathbb{1}\{i \in \text{Bin } j\} \times F_t^{Active}/A_{t-1}^{Active}$$

$$+ \sum_{j \notin \{250\}} \gamma_j^{Passive}\, \mathbb{1}\{i \in \text{Bin } j\} \times F_t^{Passive}/A_{t-1}^{Passive} + X'_{i,t}\xi + \varepsilon_{i,t}$$



$$\Delta\text{CAPM}\ \hat{\beta}_{i,t}^{21day} = \alpha_i + \alpha_t + \mathbb{1}\{i \in \text{Small}]\} \times \text{Net Flow}_t \times \text{BMI}_{i,t-1}$$
$$+ \mathbb{1}\{i \in \text{Top-50}]\} \times \text{Net Flow}_t \times \text{BMI}_{i,t-1} + \ldots + \varepsilon_{i,t}$$



Passive flows × BMI explains cross-section and time-series in simulation

back

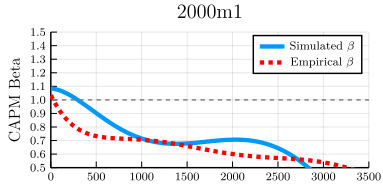
- Simulate monthly distribution of single-factor CAPM $\hat{\beta}$s from two-factor model $\tilde{R}_{i,t} = \alpha_{i,t} + \beta_{i,t}^{PC1}\lambda_t^{PC1} + \beta_{i,t}^{Flow}\lambda_t^{Flow} + u_{i,t}$

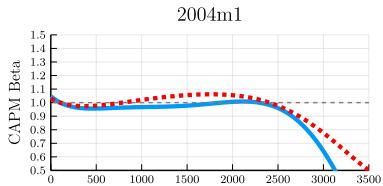
$$\text{CAPM } \hat{\beta}_{it} = \frac{w_t' b_t \Sigma_f b_t' e_i + w_{i,t}\sigma_{u,i}^2}{w_t' b_t \Sigma_f b_t' w_t + w_t' \Sigma_u w_t}, \quad \widehat{\Sigma}_{f,t} = \begin{pmatrix} \widehat{\sigma}_{z,t}^2 & \widehat{\rho}_{zf,t}\widehat{\sigma}_{z,t}\widehat{\sigma}_{f,t} \\ \widehat{\rho}_{zf,t}\widehat{\sigma}_{z,t}\widehat{\sigma}_{f,t} & \widehat{\sigma}_{f,t}^2 \end{pmatrix}$$

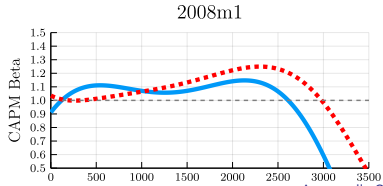
- Fundamental factor is PC1 of
  - industrial production, treasury bill, unemployment, oil price, UMichican sentiment, CPI.
- Flow factor is either active or passive net flows and exposure is BMI

Model Calibration	RMSE		Rank Correlation		Monthly
	Passive Flows	Active Flows	Passive Flows	Active Flows	Obs.
Baseline	**0.162**	**0.294**	**0.49**	**-0.02**	**249**
Full Sample $\widehat{\Sigma}_f$	0.161	0.273	0.48	0.05	249
Fixed Weights ($w_{it} = w_{i1990m5}\ \forall i$)	0.159	0.286	0.49	-0.02	249
Flow Factor Off ($\sigma_f = 0$)	0.531	0.531	-0.07	-0.07	249

Notes: This table reports the average root mean square error (RMSE) and Spearman rank correlation between simulated and empirical CAPM $\hat{\beta}$s from January 1998 to September 2018. Each month, we simulate conditional means of CAPM $\hat{\beta}$ across



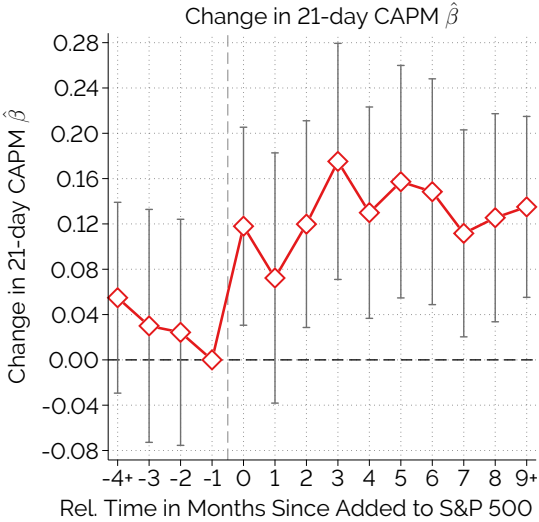


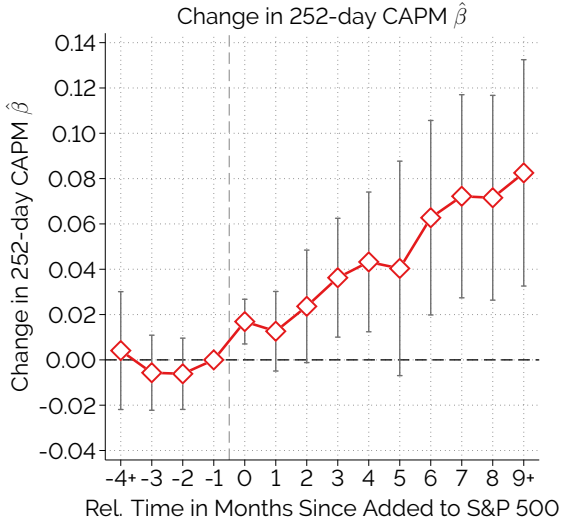


Appendix 31

Diff-in-diff at S&P 500 inclusion: CAPM $\hat{\beta}$ increases

◂ Beta Summary ◂ Inv DiD





Does managers' perceived cost of capital increase with BMI?

◂ Data ◂ PCOC

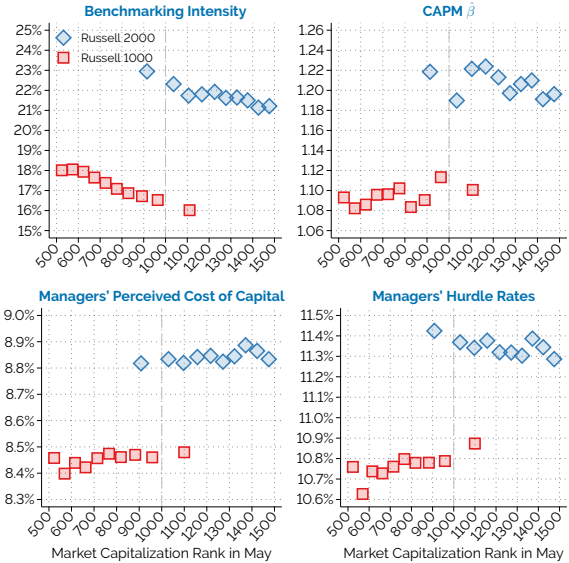
- Gormsen & Huber (2024) predict **perc. cost of capital** and **hurdle rates**
  - Lasso regression using 153 characteristics
  - External validity: Duke CFO survey (Graham, 2022

5 Add. sources of **perceived cost of equity**

- Stock analysts
  1. I/B/E/S
  2. Morningstar
  3. Value Line

  NB GH ('24) and Décaire & Graham ('25) show stock analysts' and managers' perceived cost of equity highly correlated

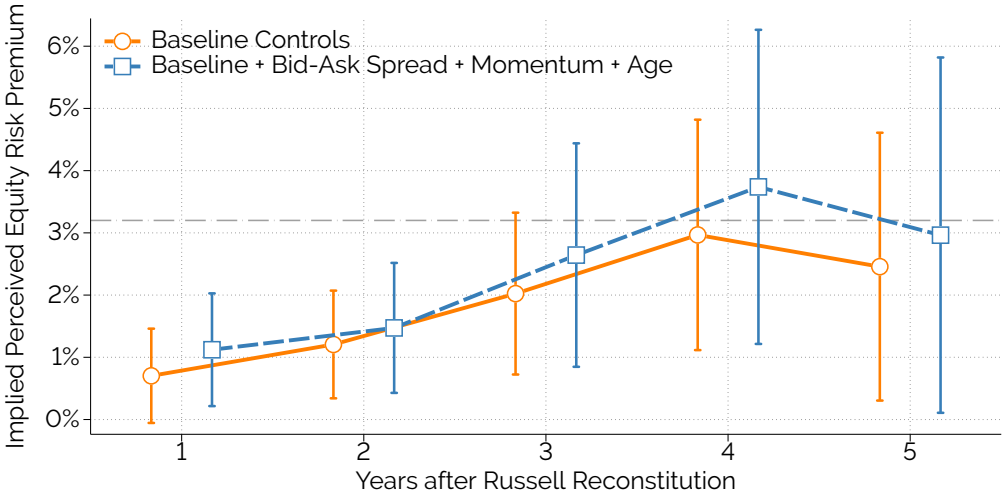
- Regulators
  4. State's Public Utility Commissions
  5. Surface Transportation Board



▸ Validation in Duke CFO Survey

Estimation in difference specification yields similar result

back



$$\Delta \text{Perceived Cost of Capital}_{i,t+h} = \delta_{j,t} + \lambda \, \widehat{\Delta \text{CAPM } \beta_{i,t+1}} + X'_{i,t}\zeta + \epsilon_{i,t+h}$$

Do firm fundamentals change with BMI?

◂ β DiD ◂ β Summary ◂ LP-RF ◂ Alt. Expl.

Most of the variation in BMI we exploit comes from benchmark inclusion...
but many things could change with benchmark inclusion?

- Does risk **exposure change** with benchmark inclusion?
  - Peer companies CAPM β ▸ go to
  - Firm-level risk measures from (Hassan et al, '19) ▸ go to
- Do changes in **financial constraints** correlate with BMI changes?
  - Text-based measures of financial constraints (Hoberg & Maksimovic, '15) ▸ go to
  - CDS spreads and CDS βs ▸ go to
- Do changes in **corporate governance** correlate with BMI changes?
  - Governance scores from S&P, Sustainalytics, and Refinitiv ▸ go to

⇛ **No evidence that risk, financial constraints, or governance changes with BMI**

Firm-level risk does not change with with BMI changes

back

(in σ units)	(1) Δ Risk	(2) Δ P. Risk	(3) Δ P. Risk Economic	(4) Δ P. Risk Security	(5) Δ P. Risk Technology	(6) Δ P. Risk Trade
Δ BMI	-0.020	-0.006	-0.003	0.003	-0.009	-0.004
	(0.014)	(0.014)	(0.012)	(0.011)	(0.011)	(0.016)
Log(Market Cap.)	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Adj. R^2	0.03	0.01	0.02	0.01	0.01	0.02
Observations	9,368	9,264	9,190	9,138	9,093	9,093

$$\Delta\text{Firm-level Risk}_{i,t} = \alpha_t + \gamma\Delta\text{BMI}_{i,t} + \log(\text{Market Cap.}) + \varepsilon_{i,t}$$

Peer firms CAPM β comove, but not when BMI changes

back

	(1) Firm's Δ CAPM β^E	(2) Firm's Δ CAPM β^E	(3) All Peers' Δ CAPM β^E	(4) Peer n=1 Δ CAPM β^E	(5) Peer n=2 Δ CAPM β^E	(6) Peer n=3 Δ CAPM β^E
Δ CAPM β^{Peer}	0.218***					
	(0.027)					
Δ BMI		0.511**	0.042	0.064	0.085	-0.023
		(0.176)	(0.053)	(0.092)	(0.081)	(0.083)
Log(Market Cap.)	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Adj. R^2	0.20	0.30	0.13	0.12	0.12	0.14
Observations	16,021	5,302	16,416	5,641	5,514	5,261

$$\Delta\text{CAPM } \beta_{j,t}^{\text{Peer}} = \alpha_t + \gamma \Delta\text{BMI}_{i,t}{}^{\text{Firm}} + \log(\text{Market Cap.}) + \varepsilon_{j,t}$$

Measures of firm-level risk

CDS spreads and CDS CAPM β do not change with BMI

back

	(1) Δ CDS Spread	(2) Δ CDS Spread	(3) Δ CDS Beta	(4) Δ CDS Beta
Δ BMI	-0.054	-0.063	-0.020	-0.031
	(0.056)	(0.049)	(0.040)	(0.032)
Momentum (Cum. Ret.)		-0.188***		-0.090
		(0.043)		(0.059)
Log(Market Cap.)	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Adj. R^2	0.14	0.16	0.00	0.01
Observations	781	781	614	614

$$\Delta\text{CDS Spreads}_{i,t} = \alpha_t + \gamma\ \Delta\text{BMI}_{i,t} + \log(\text{Market Cap.}) + \nu_{i,t}$$

Governance scores do not change with BMI

back Alternative Mechanisms

(in σ units)	(1) Δ SP G-Score	(2) Δ Sus. G-Score	(3) Δ Ref G-Score	(4) Δ SP ESG	(5) Δ Sus. ESG	(6) Δ Ref. ESG
Δ BMI	-0.066	0.004	0.006	-0.033	0.005	-0.016
	(0.072)	(0.023)	(0.020)	(0.059)	(0.019)	(0.018)
Log(Market Cap.)	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Adj. R^2	0.03	0.09	0.03	0.04	0.17	0.05
Observations	419	2,370	4,123	417	2,449	4,137

$$\Delta\text{G-Score}_{i,t} = \alpha_t + \gamma\ \Delta\text{BMI}_{i,t} + \log(\text{Market Cap.}) + \nu_{i,t}$$

Financial constraints do not correlate with BMI changes

(in σ units)	(1) Δ Inv. Delay	(2) Δ Inv. Delay & Equity Issue	(3) Δ Inv. Delay & Debt Issue	(4) Δ Inv. Delay & Private Issue	(5) Δ Inv. Delay & Equity (LW, '24)	(6) Δ Inv. Delay & Debt (LW, '24)
Δ BMI	-0.011	0.000	-0.017	-0.011	-0.010	-0.018
	(0.008)	(0.010)	(0.019)	(0.013)	(0.011)	(0.013)
Log(Market Cap.)	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Adj. R^2	0.01	0.01	0.02	0.00	0.01	0.01
Observations	5,688	5,686	5,669	5,713	7,674	7,624

$$\Delta\text{Investment Delay}_{i,t} = \alpha_t + \gamma\Delta\text{BMI}_{i,t} + \log(\text{Market Cap.}) + \varepsilon_{i,t}$$

Back

BMI predicts investment after correcting for noise in Tobin's q

back

	Dep. var.: Investment rate (CAPX/PPE)				
	OLS		IV: Industry q		EW (2002)
BMI (in %)	-0.147***	-0.105***	-0.099***	-0.070***	-0.067***
	(0.021)	(0.022)	(0.022)	(0.020)	(0.014)
Tobin's q^{tot}	3.434***	3.038***	5.043***	4.573***	6.078***
	(0.155)	(0.205)	(0.327)	(0.435)	(0.157)
Leverage	0.008*	0.012**	0.018***	0.006	0.001
	(0.005)	(0.006)	(0.006)	(0.006)	(0.005)
Cash Flow	-0.319***	-0.119	-0.291***	-0.152	-0.190**
	(0.061)	(0.117)	(0.055)	(0.105)	(0.088)
Firm FE		✓		✓	✓
Year × Rank-Bin FE	✓	✓	✓	✓	✓
FS F-stat.			143.16	33.95	
Within R^2	0.21	0.12	0.17	0.10	0.27
Observations	36,277	36,129	35,690	35,543	36,129

1 SD higher BMI is associated with a 7.4% *lower* investment rate

back

	Dependent variable: (XRD + 0.3 SGA)$_{t+1}$ / K$_t^{INT}$ (in %)				
BMI (in %)	-0.192***	-0.316***	-0.167***	-0.058***	-0.058***
	(0.053)	(0.050)	(0.045)	(0.020)	(0.020)
Tobin's q^{tot}				3.053***	2.975***
				(0.204)	(0.169)
Linear Time Trend	-0.674***				
	(0.063)				
Leverage				-0.003	-0.004
				(0.006)	(0.006)
Other Controls				✓	✓
Firm FE	✓		✓	✓	✓
Year × Size FE		✓	✓	✓	
Industry × Year × Size FE					✓
Adj. R^2	0.59	0.11	0.62	0.69	0.75
Mean Dep. Var.	22.0	22.2	22.0	21.9	22.0
SD BMI	9.7	9.8	9.7	9.7	9.6
Observations	37,771	38,390	37,769	36,545	35,192

Appendix 42

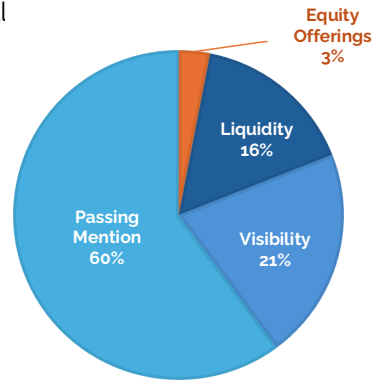
Executives Commenting on Russell Index Inclusion

◂ Quotes ◂ ICC

- Analyze all 163 earnings call transcripts that mention Russell index inclusion from 2008 to 2024

- Assign transcripts sequentially into one of four categories:

1. **Equity Offerings:** Discussion of an at-the-market offering, seasoned equity offering, shelf takedown, or other equity raise
2. **Liquidity:** Mention increased stock liquidity or trading volume
3. **Visibility:** Mention increased visibility or a broader investor base
4. **Passing Mention:** A brief note of inclusion, no further context



Modal comment:

"The first half of this year has been extremely productive, and we are driving continued progress, sustainable and profitable growth as a leader and the disruptor of the B2B e-commerce technology solution. ***We were honored to be added to the Russell 2000 Index through their recent reconstitution.*** *Now I will turn the call over to Iman to provide more color and – on our operational highlights."*

"Top line data from our PEDFIC-1 Phase III trial is expected at the end of 2019 or early 2020. ***And finally, we were selected for inclusion in the Russell 2000 Index Russell as part of the annual reconstitution of the Russell stock indexes.*** *And with that, let me turn the call back over to Ron for some concluding remarks."*

Which $\hat{\beta}$ do managers use? Daily, weekly, monthly?

◂ β res ◂ PCOC res

- **FactSet**: 252 **daily** returns
- **Bloomberg**: 104 **weekly** returns
- **S&P Capital IQ**: 156 **weekly** returns
- **Value Line**: 260 **weekly** returns
- **Yahoo Finance**: 36 **monthly** returns (change to 60 months recently)
- **Morningstar**: 36 **monthly** returns
- **Refinitiv**: 60 **monthly** returns

ROIC Target (in %) in Executive Compensation Plan			
β using 36 monthly obs.	1.772**		
	(0.641)		
β using 156 weekly obs.		3.726***	
		(0.886)	
β using 252 daily obs.			5.517***
			(1.019)
Firm FE	✓	✓	✓
Year FE	✓	✓	✓
Adj. R^2	0.69	0.69	0.72
Within-R^2	0.01	0.02	0.04
Observations	1,340	1,340	1,340

➠ Measurement error → low R^2 and attenuation bias

Managers reference CAPM when talking about cost of equity

back

*"If you use some of the tools I learned in my MBA class, like the **capital asset pricing model**, they did teach that back in the 80s by the way, so it's been around for a while. I think our cost of equity is around 10%"*

— CFO, Qorvo Inc. (Q4 2015)

*"This is our way of calculating the cost of equity. So this is very textbook like **CAPM** type of methodology. We start with the risk-free rate [...]. Then we look at the group's beta."*

— CFO, Talanx (Q4 2019)

*"If you don't push your ROE target up when interest rates go up, and that's for every company to decide, it depends if you believe in the **capital asset pricing model** or not. I won't take you back to school. [...] I think you should [expect a higher ROE]."*

— CFO, Intact Financial (Q3 2024)

*"So the investment is about $240 million. We expect to project finance the deal so we will be, I think, around $130 million equity investment. [...] I'm not going to get in to what our cost of capital is but most people can probably — I mean everybody knows **CAPM** and can kind of back into what it is."*

— CEO, Vistra Energy (Q1 2017)

*"From a cost of capital perspective, we update our cost of capital estimates monthly. So yes, there are changes that go on a monthly basis looking at interest rates and looking at local country risk, as well as betas and all of the things that go into kind of the **CAPM** model."*

— CEO, American Tower Corp. (Q4 2013)

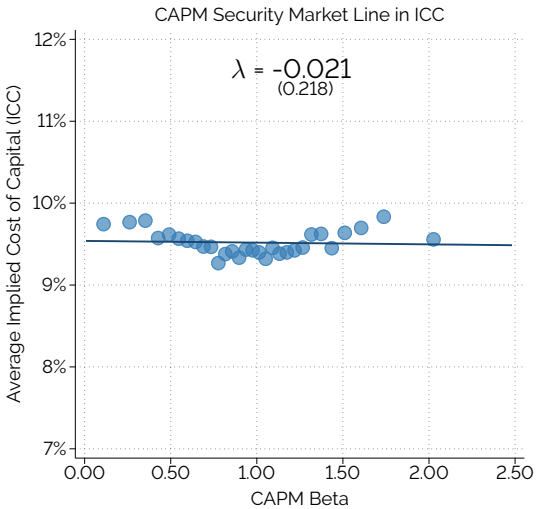
*"We haven't historically disclosed our cost of equity, but it's actually fairly straightforward calculation from a **CAPM** model, so in terms of where we're coming out. But I think from that, you're going to get around a 10%-ish type range cost."*

— CEO, E-Trade (Q1 2016)

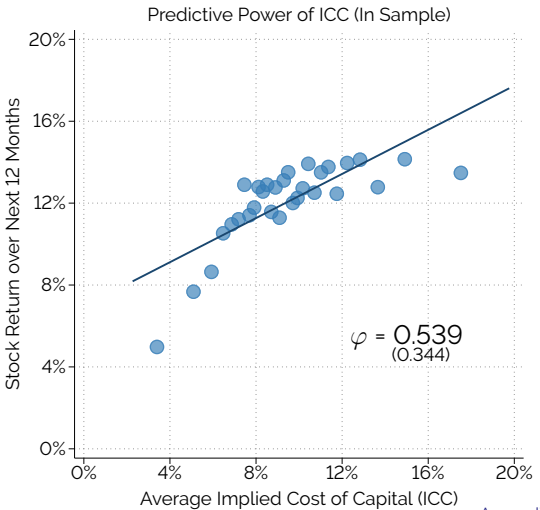
CAPM security market line is flat in implied cost of capital

back

$$ICC_{i,t} = \alpha_t + \lambda\,\hat{\beta}_{i,t} + \varepsilon_{i,t}$$



$$Return_{i,t\to t+12} = \alpha_t + \varphi\ ICC_{i,t} + \epsilon_{i,t}$$



Appendix 46

Managers perceived cost of capital increases after exog. increase in BMI

back

	Perceived Cost of Capital$_{t+1}$ (in p.p.)				Managers' Hurdle Rate$_{t+3}$ (in p.p.)			
	RF		IV		RF		IV	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ Benchmarking Intensity$_t$ (in p.p.)	0.016**	0.014**			0.025**	0.025**		
	(0.007)	(0.005)			(0.010)	(0.009)		
$\widehat{\text{CAPM}}\ \beta^A_{t+1}$			3.422**	3.282**			5.232**	5.489**
			(1.307)	(1.183)			(2.318)	(2.252)
Baseline Controls	✓	✓	✓	✓	✓	✓	✓	✓
Additional Controls		✓		✓		✓		✓
Fixed Effects								
Year	✓		✓		✓		✓	
Year $\times$ Industry		✓		✓		✓		✓
FS F-stat.			12.89	16.36			12.86	16.88
Adj. R^2	0.32	0.52			0.07	0.34		
Observations	7,733	7,696	7,733	7,696	6,684	6,649	6,684	6,649

Stock analysts responds to exog. increase in BMI by raising cost of equity

back

	Δ Morningstar Cost of Equity (in p.p.)				Value Line Safety Rank × 1.5 p.p.				Δ I/B/E/S Expected Return (in p.p.)			
	RF		IV		RF		IV		RF		IV	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Δ BMI	0.023***	0.020**			0.030**	0.029**			0.101*	0.084*		
	(0.008)	(0.009)			(0.010)	(0.011)			(0.055)	(0.045)		
$\widehat{\Delta \text{ CAPM } \beta}$			4.867**	4.117**			4.425**	3.953*			12.336*	10.402*
			(2.276)	(1.976)			(1.673)	(1.956)			(6.243)	(5.189)
Baseline Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Additional Controls		✓		✓		✓		✓		✓		✓
Fixed Effects												
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
FS F-stat.			9.9	11.9			25.1	18.1			87.3	88.4
Observations	5,721	5,381	5,514	5,389	1,866	1,866	1,809	1,809	4,692	4,692	4,692	4,692

Regulators approve higher cost of equity for firms with higher BMI

back

	Dependent Variable: Authorized Cost of Equity $- r^f$							
	Public Utilities				Railroads			
	RF		IV		RF		IV	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
BMI (in %)	0.069***	0.071***			0.481***	0.454***		
	(0.011)	(0.011)			(0.099)	(0.088)		
$\widehat{\text{CAPM } \beta^E}$			6.064***	6.189***			6.462***	6.481***
			(1.386)	(1.375)			(0.331)	(0.361)
DCF implied Cost of Equity $- r^f$		0.281***		0.170**		0.654**		-0.035
		(0.051)		(0.083)		(0.327)		(0.072)
Fixed Effects								
Utility × State FE	✓	✓	✓	✓				
Adj. R^2	0.26	0.43			0.48	0.60		
FS F-stat.			42.5	45.4			23.1	25.4
Observations	1,052	1,052	1,052	1,052	21	21	21	21

Placebo test with cost of debt

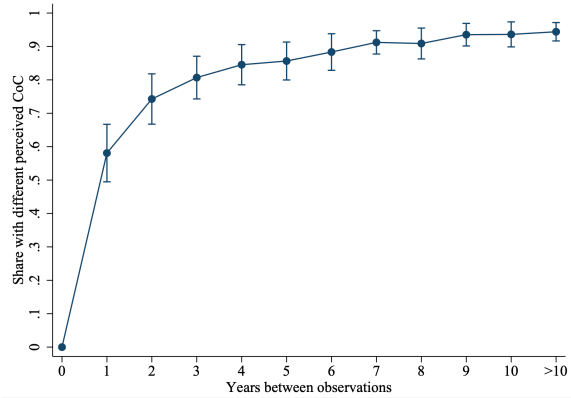
Regulators do not approve higher cost of debt for firms with higher BMI

back

	Requested			Authorized		
	Return on Debt — R^f			Return on Debt — R^f		
Benchmarking Intensity (in %)	0.016	0.008	0.009	-0.002	-0.013	-0.020
	(0.012)	(0.011)	(0.011)	(0.014)	(0.013)	(0.015)
BBB Option-Adjusted spread		0.264***	0.259***		0.310***	0.296***
		(0.048)	(0.049)		(0.053)	(0.057)
Requested E/(D+E)		0.075***	0.071***			
		(0.016)	(0.018)			
Authorized E/(D+E)					0.072***	0.066***
					(0.018)	(0.016)
Constant	2.218***	-1.979*		3.502***	-0.487	
	(0.267)	(0.753)		(0.295)	(0.878)	
Utility-by-State Fixed Effect			✓			✓
Adj. R^2	0.00	0.14	0.38	0.00	0.13	0.39
Observations	1,381	1,381	1,347	1,022	1,022	987

Pass-thru from managers' perc. cost of capital to hurdle rates

◂ PCoC Res. ◂ Inv. Res.



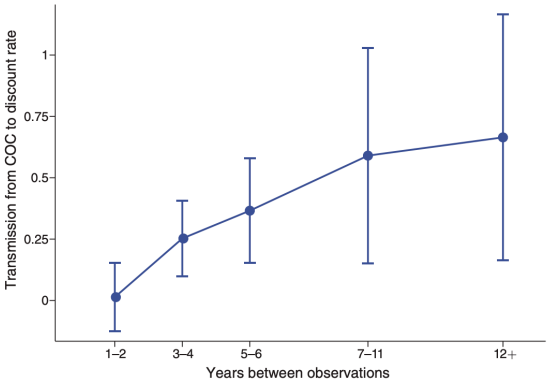


FIGURE 4. THE TRANSMISSION FROM THE COST OF CAPITAL TO DISCOUNT RATES OVER TIME

- Managers update Perceived Cost of Capital updated frequently
- Pass-through from Perceived Cost of Capital to Hurdle Rate sluggish

Predicted data of Gormsen & Huber ('24) performs well vs. Duke CFO Survey

back

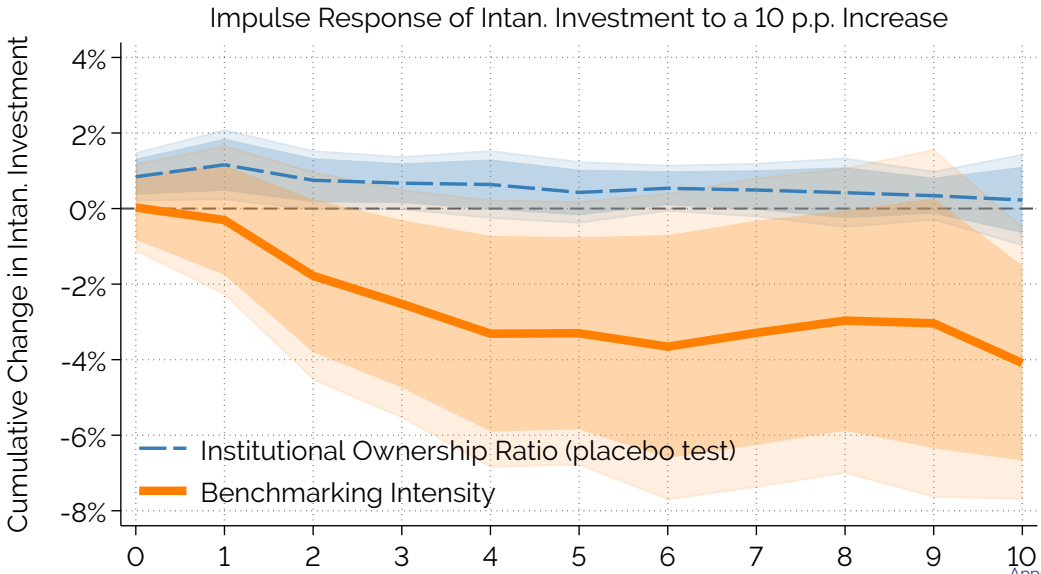
Table A6
Comparison of Predicted Data and Duke CFO Data

Columns (1) and (2) report regressions of the perceived cost of capital from the Duke CFO Survey on the predicted perceived cost of capital (predicted based on the conference call data). Columns (3) and (4) report regressions of discount rates (hurdle rates) from the Duke CFO Survey on the predicted discount rates (predicted based on the conference call data). The sample includes only U.S. firms. Standard errors are clustered at the firm and year level. Statistical significance is denoted by *** p<0.01, ** p<0.05, * p<0.1.

	(1) Duke CoC	(2) Duke CoC	(3) Duke discount rate	(4) Duke discount rate
Predicted CoC	0.74***	0.90***		
	(0.17)	(0.21)		
Predicted discount rate			1.02***	0.98**
			(0.38)	(0.38)
Constant	0.034**	0.021	0.027	0.031
	(0.014)	(0.018)	(0.036)	(0.037)
Observations	319	319	92	92
R-squared	0.057	0.067	0.118	0.136
FE	None	Year	None	None
Within R^2	0.057	0.057	0.12	0.11

Intan. investment falls by 4% in response to a 10 p.p. y-o-y BMI increase

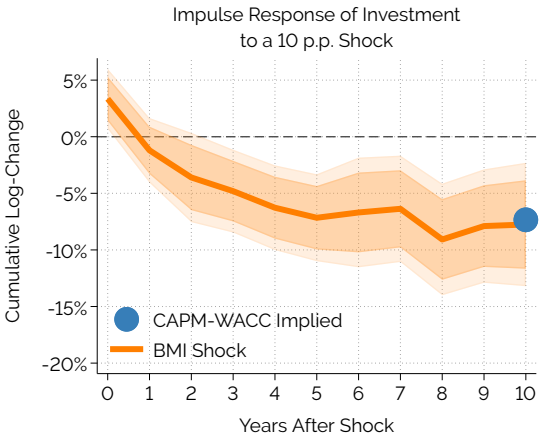
back



Appendix 53

Is dynamic response of investment consistent with cost of capital increase?

back



$$\text{Cost of Capital Semi-Elasticity} = \frac{\partial Inv}{Inv}\frac{1}{\partial r} = \frac{\widehat{d\log Inv}_{t+10}}{\underset{75\%}{E/(D+E)} \times \underset{6\%}{ERP} \times \underset{0.25}{\Delta\hat{\beta}}}$$

$$\approx -6.4$$

- Stronger effects for firms with
  - high equity share
  - long-duration investments

Appendix 54

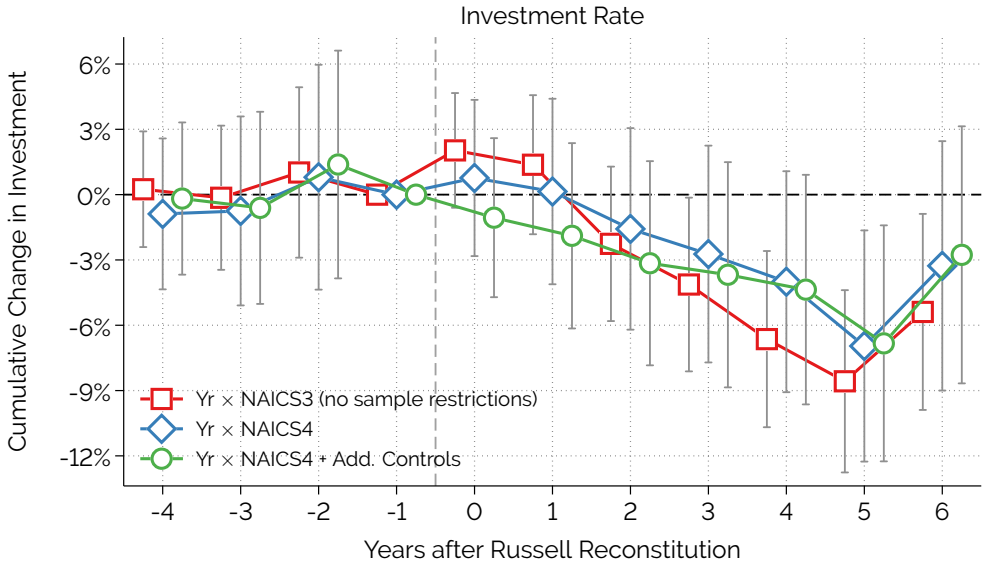
Investment decreases and payouts increase after exog. BMI increase

back

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Dep. var.: Cumulative log change in investment $100 \times (\log(Inv_{i,t+h}) - \log(Inv_{i,t}))$						
Horizon $t+h$:	$t+1$	$t+2$	$t+3$	$t+4$	$t+5$	$t+6$
Δ BMI (p.p.)	0.036	-0.500	-0.703*	-0.957**	-1.060**	-0.755*
	(0.257)	(0.398)	(0.343)	(0.448)	(0.488)	(0.418)
Log(Mkt. Cap) and Log(Bid-Ask Spread)	✓	✓	✓	✓	✓	✓
Banding Variables	✓	✓	✓	✓	✓	✓
Window width around Russell 1000 cutoff	300	300	300	300	300	300
Adj. R^2	0.07	0.11	0.11	0.11	0.10	0.08
Observations	8,253	7,829	7,043	6,320	5,629	5,021
Panel B: Dep. var.: Change in net payout ratio $100 \times (\text{Net Payout Ratio}_{i,t+h} - \text{Net Payout Ratio}_{i,t})$						
Horizon $t+h$:	$t+1$	$t+2$	$t+3$	$t+4$	$t+5$	$t+6$
Δ BMI (p.p.)	0.069	0.111	0.144	0.143*	0.184**	0.080*
	(0.059)	(0.078)	(0.087)	(0.070)	(0.080)	(0.042)
Log(Mkt. Cap) and Log(Bid-Ask Spread)	✓	✓	✓	✓	✓	✓
Window width around Russell 1000 cutoff	300	300	300	300	300	300
Adj. R^2	0.02	0.05	0.06	0.06	0.06	0.06
Observations	8,080	7,678	6,903	6,182	5,496	4,915

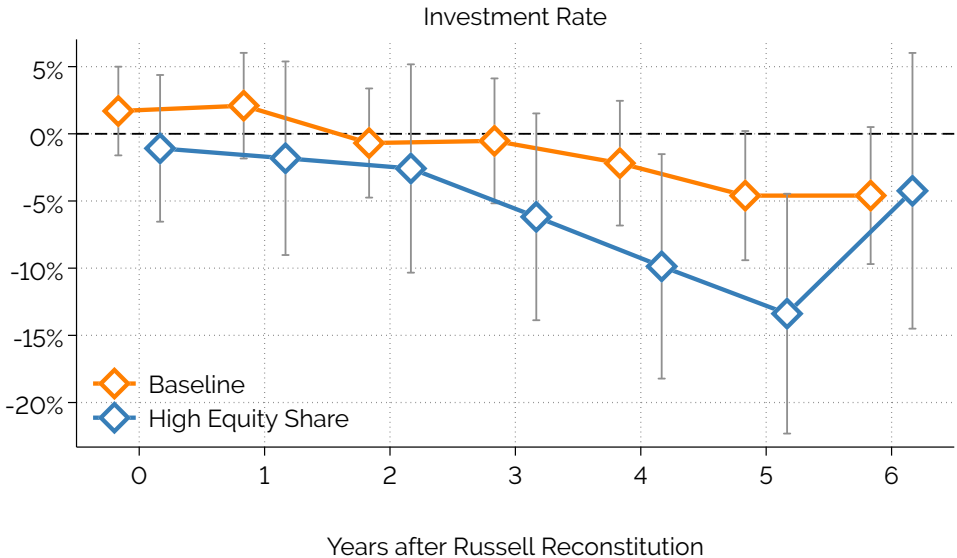
Additional controls and fixed effects

back



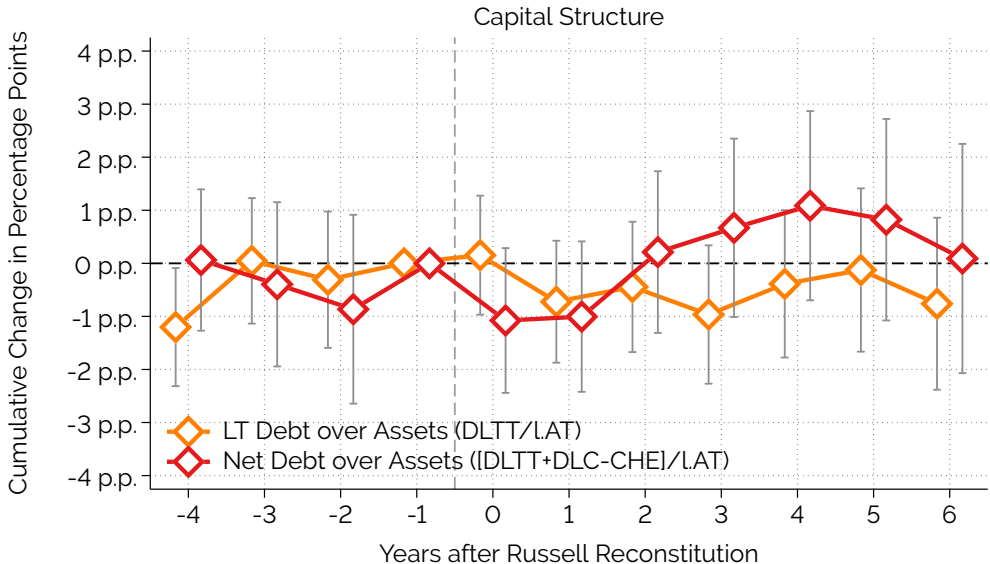
Results are stronger for firms with higher equity share

back



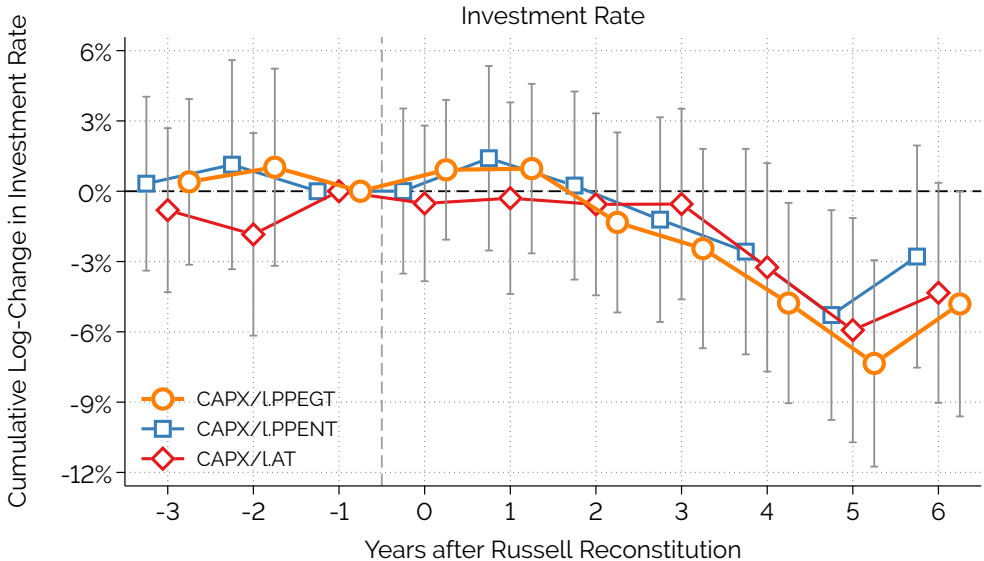
Capital structure does not change after Russell recon.

back



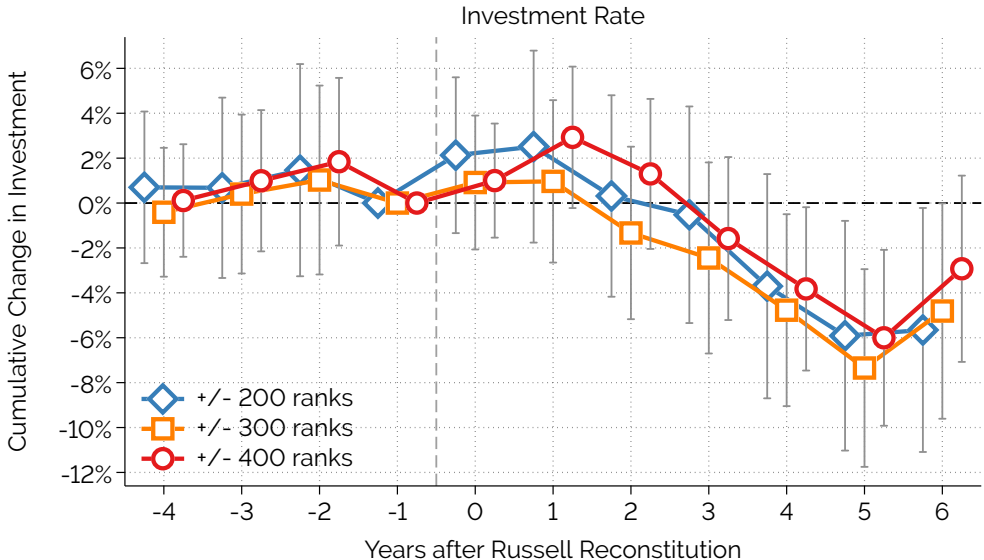
Similar results using other investment rate definition

back

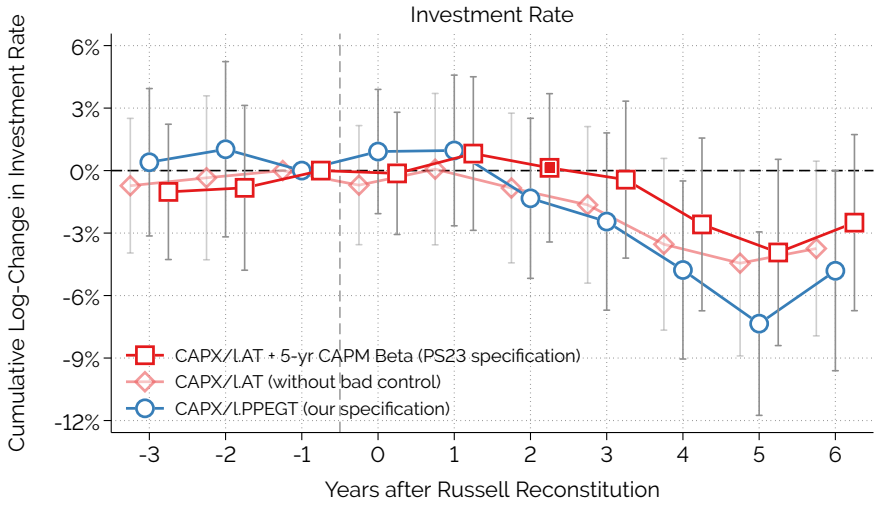


Similar results using different windows around cutoffs

back



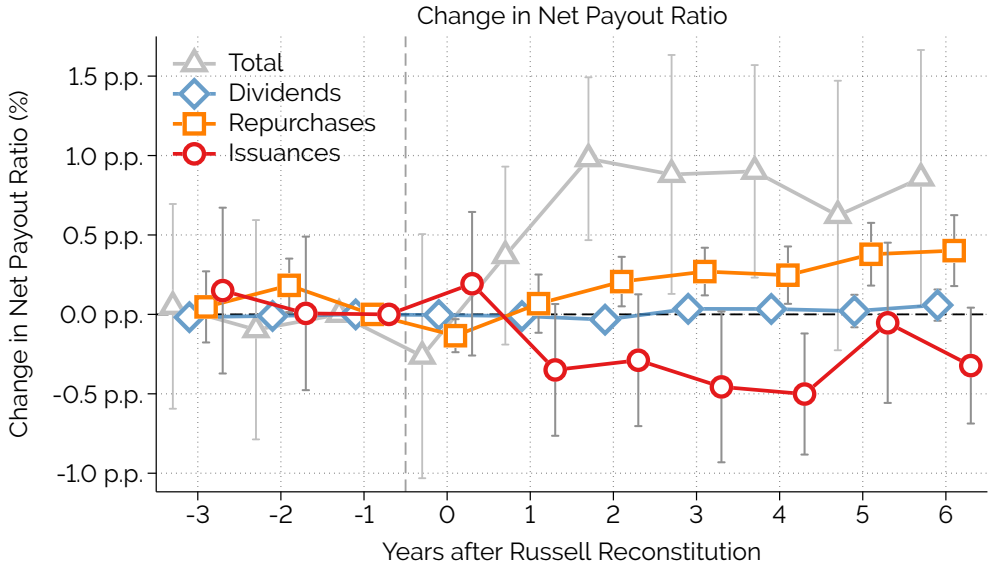
Reconciliation with Pavlova & Sikorskaya (2023) back



- Carl Sagan: "*Absence of evidence is not evidence of absence.*" (see also McCloskey & Ziliak, 1996)

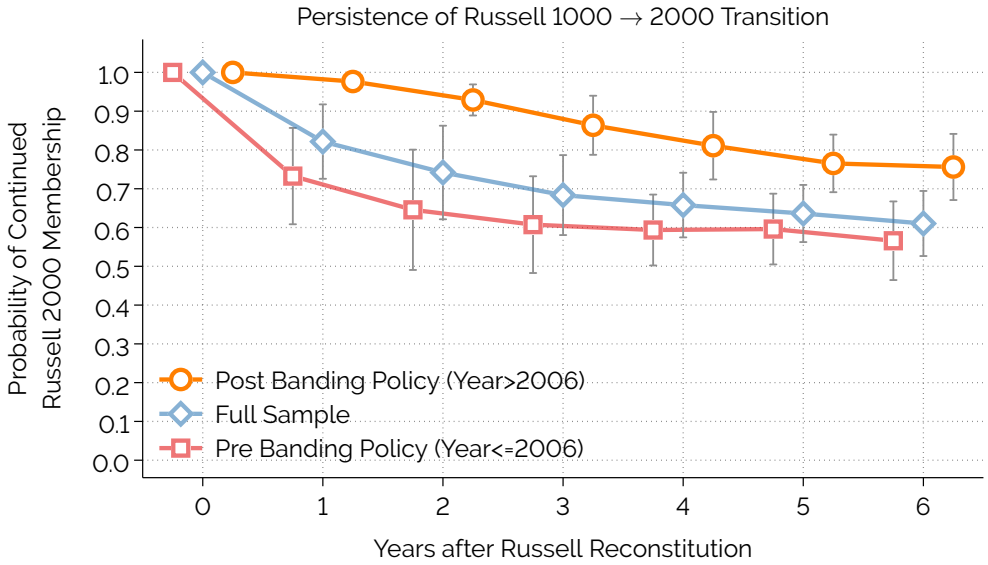
Issuance falls and repurchases increase, dividends unchanged

back



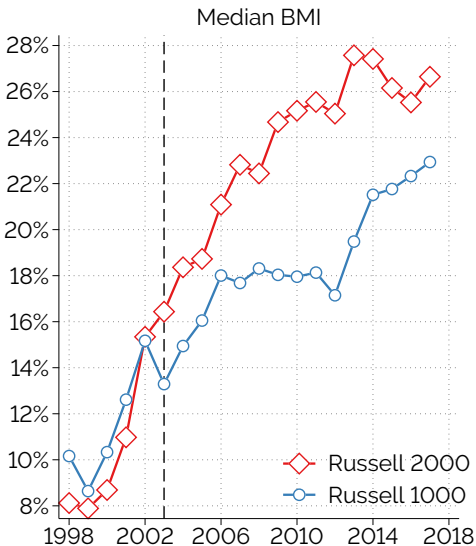
Transition is persistence, especially post banding policy

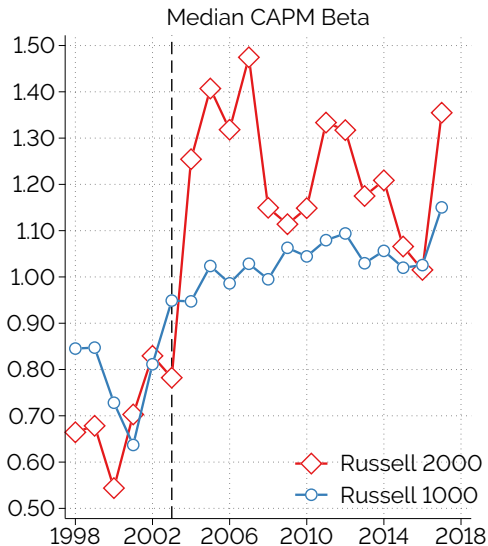
◂ Inv. ◂ Beta



Effects are concentrated post-2003 when $BMI_{2000} > BMI_{1000}$

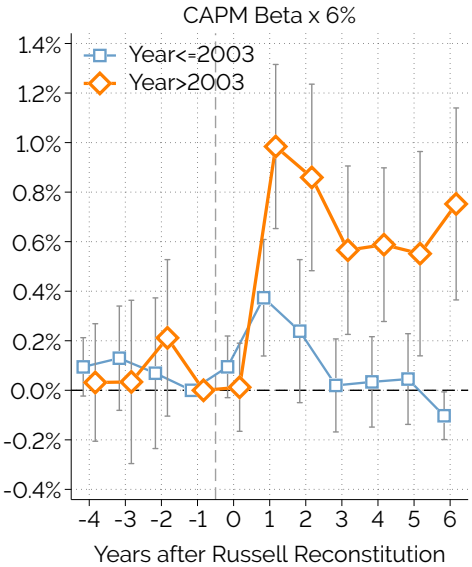
Inv.

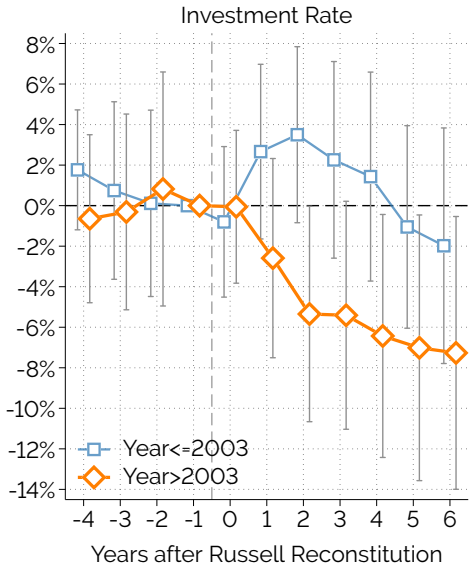




Effects are concentrated post-2003 when $BMI_{2000} > BMI_{1000}$

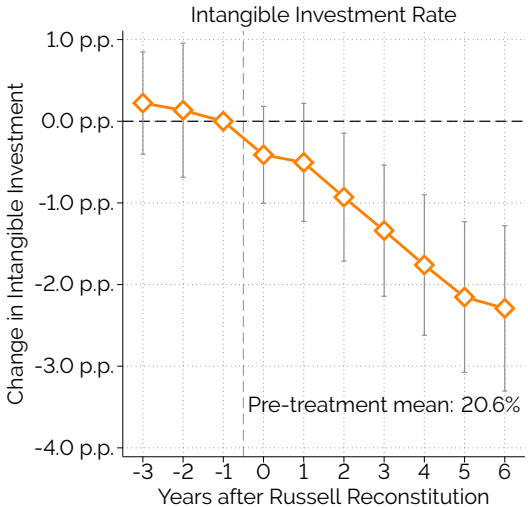
Inv.

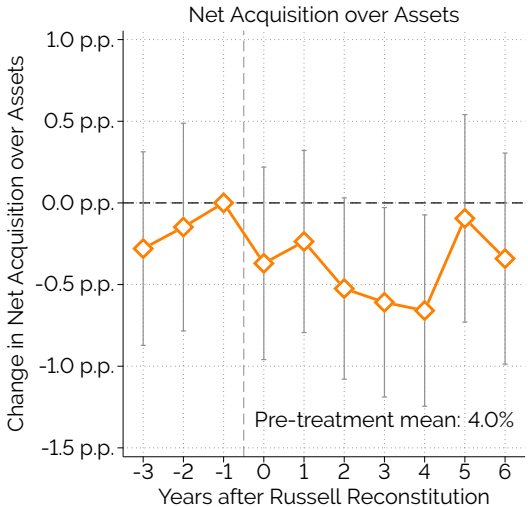




Intangible investment falls by 11%, acquisitions fall by 9%

back



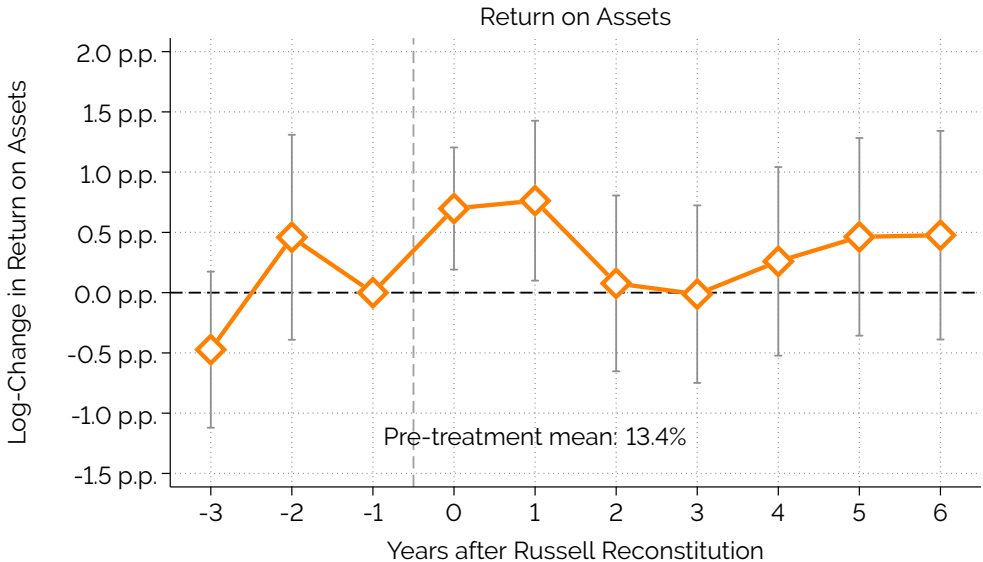


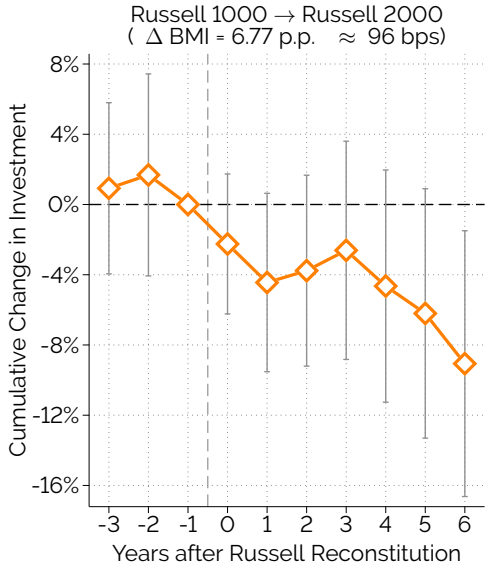
- Definition of intangibles from Peters & Taylor (2016): XRD + 0.3 SGA

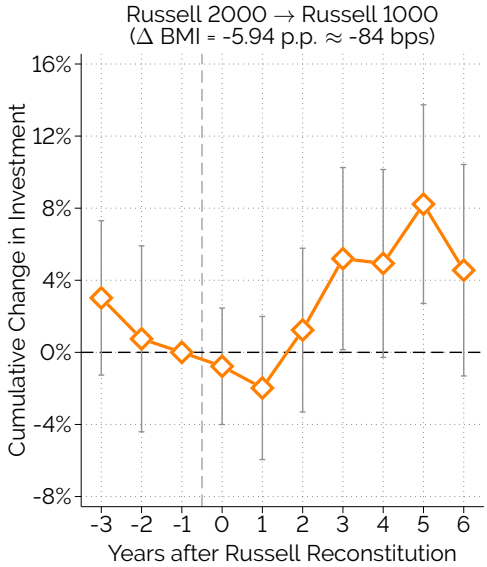
see also Hulten and Hao (2008), Eisfeldt and Papanikoloau (2014), and Zhang (2014)

Profitability / Return on Assets is stable

back

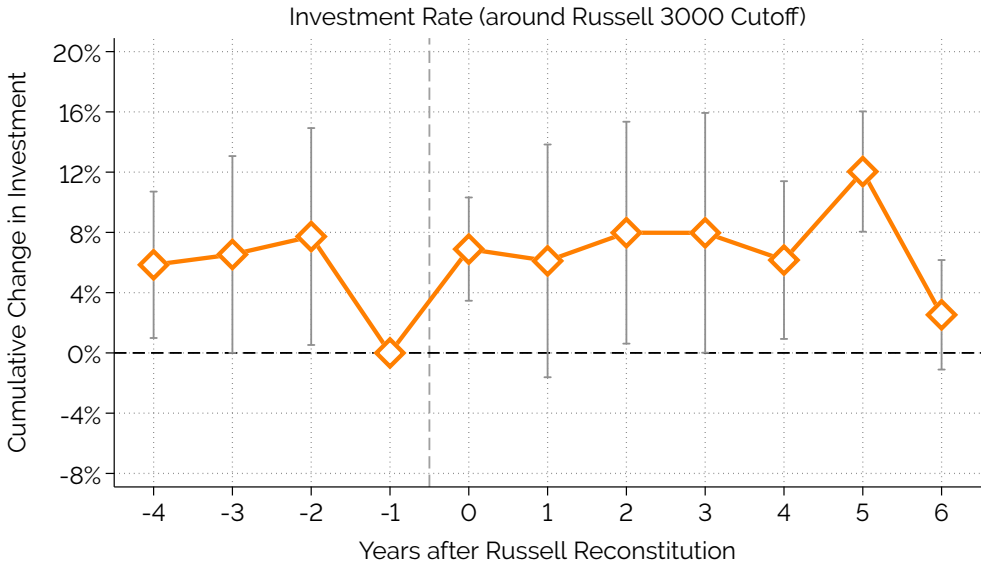



Russell 1000 → Russell 2000
(Δ BMI = 6.77 p.p. ≈ 96 bps)
Cumulative Change in Investment
8%
4%
0%
-4%
-8%
-12%
-16%
-3 -2 -1 0 1 2 3 4 5 6
Years after Russell Reconstitution


Russell 2000 → Russell 1000
(Δ BMI = -5.94 p.p. ≈ -84 bps)
Cumulative Change in Investment
16%
12%
8%
4%
0%
-4%
-8%
-3 -2 -1 0 1 2 3 4 5 6
Years after Russell Reconstitution

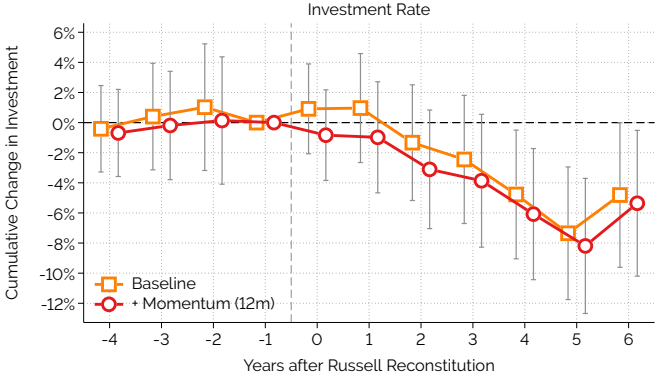
Firms at Russell 3000 cutoff have severe CAPX pre-trends

back



Accounting for stock price momentum and selection

back



	t+0	t+1	t+2	t+3	t+4	t+5	t+6
Δ BMI	-0.123	-0.144	-0.459	-0.571	-0.898**	-1.210***	-0.792*
	(0.270)	(0.331)	(0.354)	(0.396)	(0.391)	(0.403)	(0.435)
Momentum (12m)	0.130***	0.199***	0.168***	0.113***	0.106***	0.077**	0.074*
	(0.023)	(0.035)	(0.039)	(0.041)	(0.032)	(0.036)	(0.040)
Log(Mkt. Cap) and Log(Bid-Ask Spread)	✓	✓	✓	✓	✓	✓	✓
Banding Variables	✓	✓	✓	✓	✓	✓	✓
Window width around Russell 1000 cutoff	300	300	300	300	300	300	300

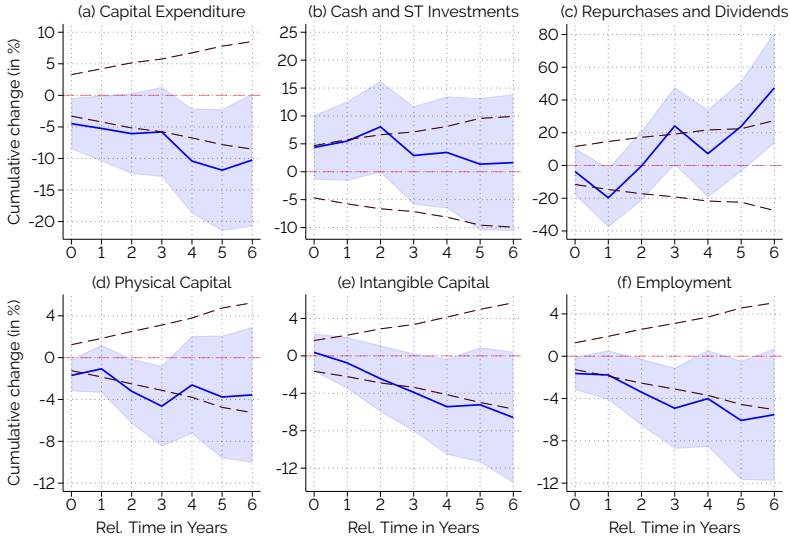
Appendix 69

Higher CAPM $\widehat{\beta}$s due to Benchmarking lead to lower investment

back

$$\Delta \log Y_{i,t+h} = \gamma^h \widehat{\Delta \text{CAPM } \beta} + \text{Ind.} \times \text{ Yr FE} + \text{Controls} + \varepsilon_{i,t+h}$$

Dose: $0.167 \times 6\% \approx 100$ *bps*



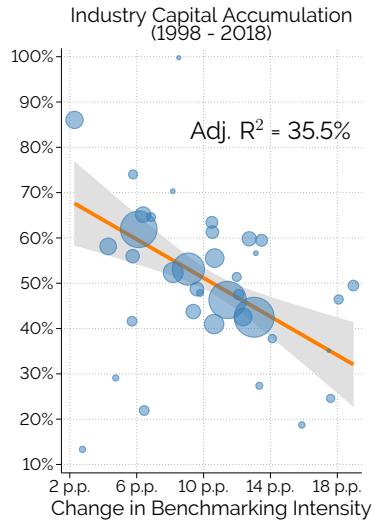
Industries with higher BMI accumulate less capital

back

- Long-difference IV regressions from 1998 to 2016

$$\Delta \text{ CAPM } \hat{\beta}_i = \delta_j + \theta \Delta \text{ BMI}_i + X_i' \zeta + \epsilon_i$$

$$\Delta \log (\text{Real Capital Stock})_i = \alpha_j + \gamma \widehat{\Delta \text{ CAPM } \beta}_i + X_i' \xi + \varepsilon_i$$



Industries with higher BMI accumulate less capital

back

- Long-difference IV regressions from 1998 to 2016

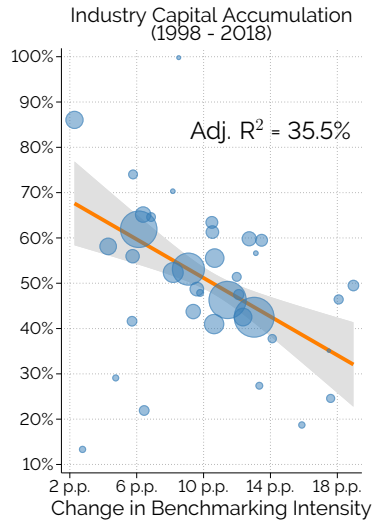
$$\Delta \text{ CAPM } \hat{\beta}_i = \delta_j + \theta \Delta \text{ BMI}_i + X_i' \zeta + \epsilon_i$$

$$\Delta \log (\text{Real Capital Stock})_i = \alpha_j + \gamma \widehat{\Delta \text{ CAPM } \beta}_i + X_i' \xi + \varepsilon_i$$

Dep. var: log (Real Capital Stock in 2016 / Real Capital Stock in 1998)				
Δ CAPM β	-0.271*	-0.329**	-0.325*	-0.328**
(1998–2016)	(0.157)	(0.129)	(0.164)	(0.157)
Pretrend Capital				-0.309***
(1980–1996)				(0.106)
Pretrend Employ.				0.256**
(1980–1996)				(0.107)
Industry Controls		✓	✓	✓
Subsector FE			✓	✓
FS F-stat.	12.95	13.79	17.60	17.37
Observations	103	103	103	103

- Δ CAPM $\widehat{\beta} = 0.3 \Rightarrow 9.9\%$ lower capital accumulation

Robust to China shock exposure



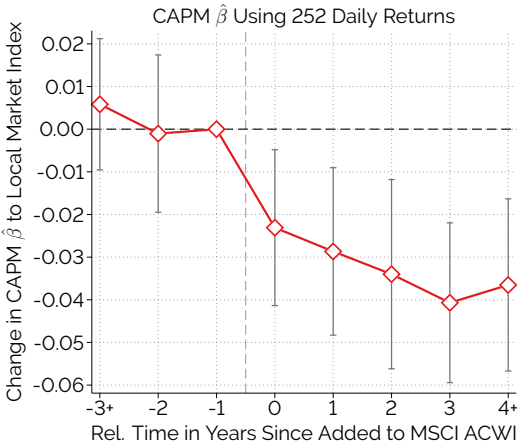
Benchmarking channel is robust to China shock (Autor et al, '13)

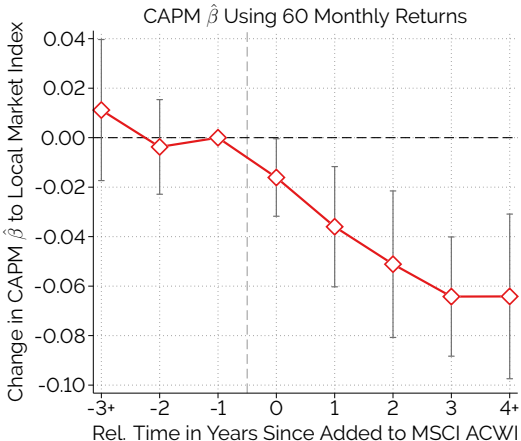
back

	Dep. var.: $\log\left(\text{Capital}^{`16}/\text{Capital}^{`00}\right)$					
ΔCAPM $\beta'^{00\to'16}$	-0.452^{+}	-0.511*	-0.592*	-0.435*	-0.453**	-0.470*
	(0.240)	(0.233)	(0.245)	(0.185)	(0.169)	(0.192)
Δ in US exposure to Chinese imports (1991–2011)		-0.157*	-0.128*		-0.078*	-0.074*
		(0.074)	(0.058)		(0.038)	(0.036)
$\log\left(\text{Emp}'^{00}\right)$			0.052			0.065^{+}
			(0.048)			(0.037)
$\log\left(\text{TFP}'^{00}\right)$			-0.200^{+}			-0.0465
			(0.115)			(0.075)
Fixed Effects						
Sector				✓	✓	✓
FS F-stat.	7.27	8.20	12.38	19.72	18.02	14.42
Weights	VA$^{`00}$	VA$^{`00}$	VA$^{`00}$	VA$^{`00}$	VA$^{`00}$	VA$^{`00}$
Observations	90	90	90	86	86	86

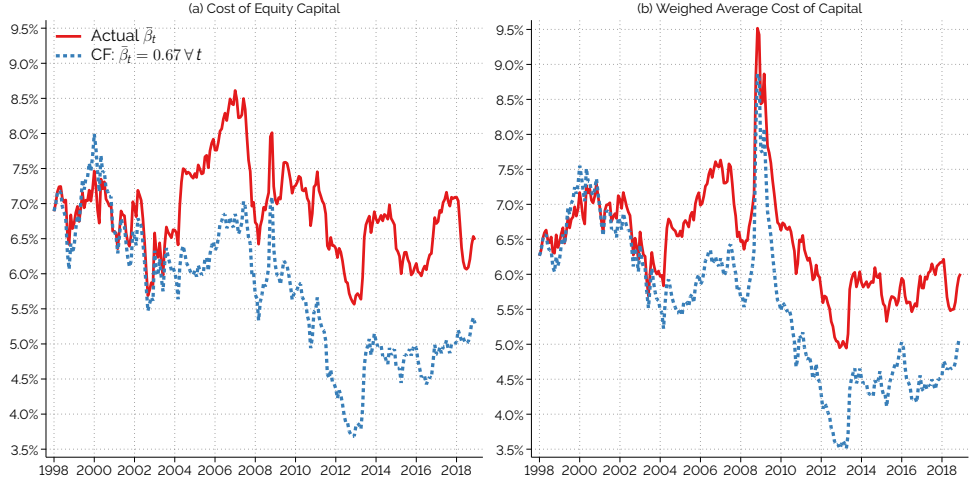
Difference-in-difference of MSCI ACWI Inclusion on local market CAPM $\hat{\beta}$

back





Counterfactual WACC, adjusted for BMI, is 150 bps lower



$$r_t^E = r_t^f + \beta_t \times \underbrace{\left(\mathbb{E}_t\left[r^{Mkt}\right] - r_t^f\right)}_{=6\%}$$

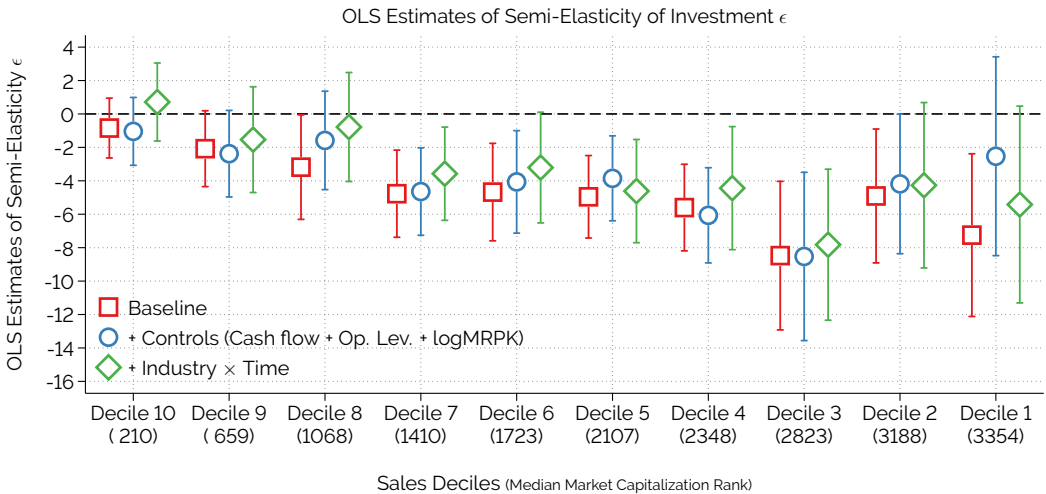
$$\text{WACC}_t = \omega_t \times r_t^E + (1-\omega_t) \times r_t^D \times (1-\tau_t)$$

Appendix 74

Smaller firms are more elastic wrt CAPM cost of equity capital

back

$$\Delta \log (Inv)_{i,j,t} = \alpha_i + \alpha_t + \phi_j^{\beta} \beta_{i,t} + \nu_{i,j,t}$$



Appendix 75

Agg. investment depends on investment share weighted distributions

back

- Let $\pi_{i,t}$ share of firm i in agg. investment, λ equity risk premium,

$$\begin{aligned}\frac{\Delta I_t}{I_{t-1}} &\approx \lambda \sum_i \pi_{i,t-1}\, \epsilon_{i,t}\, \mu_{i,t}\, \Delta\beta_{i,t} \\ &\approx \lambda \Big[\mathbb{E}_\pi[\epsilon_{i,t}]\ \mathbb{E}_\pi[\mu_{i,t}]\ \mathbb{E}_\pi[\Delta\beta_{i,t}] + \mathrm{Cov}_\pi(\epsilon_{i,t}, \Delta\beta_{i,t})\ \mathbb{E}_\pi[\mu_{i,t}] \\ &\qquad + \mathrm{Cov}_\pi(\mu_{i,t}, \Delta\beta_{i,t})\ \mathbb{E}_\pi[\epsilon_{i,t}] + \mathrm{Cov}_\pi(\epsilon_{i,t}, \mu_{i,t})\ \mathbb{E}_\pi[\Delta\beta_{i,t}]\Big]\end{aligned}$$

- Determinants of aggregate investment changes are the π-weighted distributions of
  - ϵ : semi-elasticities with respect to cost of capital
  - $\Delta\beta$: CAPM $\hat{\beta}$ changes
  - μ : share of equity in total capital

  across firms. (NB: market cap shares ω do not enter)

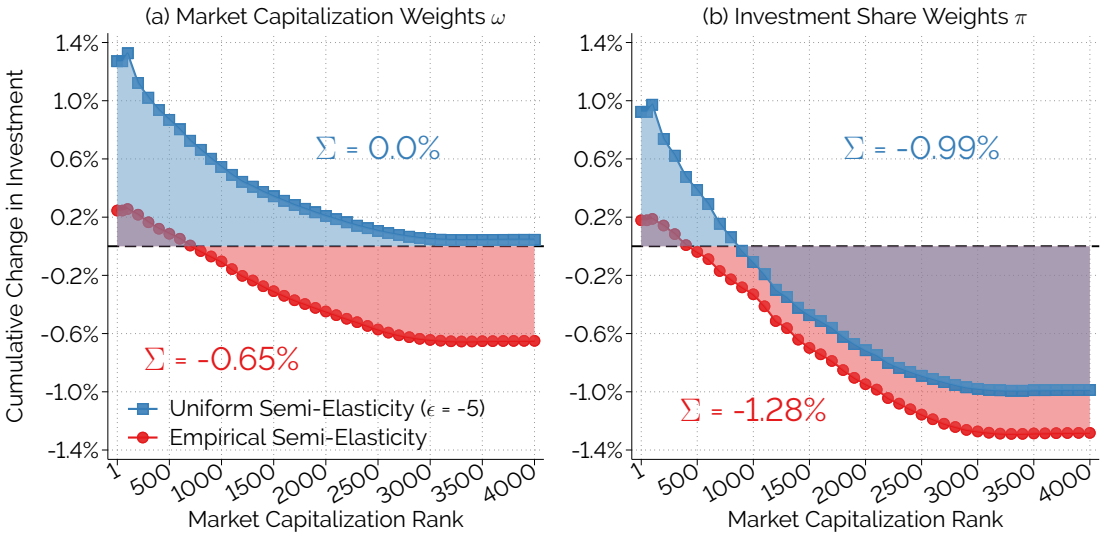
Aggregate investment falls by **1.28%** even though $\sum \omega\beta = 1$ back

Component		Estimate	Std. Err.	Share
Average Firms' Response	$\mathbb{E}_\pi[\epsilon_{i,t}]\ \mathbb{E}_\pi[\mu_{i,t}]\ \mathbb{E}_\pi[\Delta\beta_{i,t}]$	-0.067	(0.096)	0.31
+ **Sensitivity Sorting**	$\mathrm{Cov}_\pi(\epsilon_{i,t}, \Delta\beta_{i,t})\ \mathbb{E}_\pi[\mu_{i,t}]$	**-0.149*****	**(0.033)**	**0.70**
+ Equity-share Sorting	$\mathrm{Cov}_\pi(\mu_{i,t}, \Delta\beta_{i,t})\ \mathbb{E}_\pi[\epsilon_{i,t}]$	0.003**	(0.001)	-0.01
+ Pass-through Sorting	$\mathrm{Cov}_\pi(\epsilon_{i,t}, \mu_{i,t})\ \mathbb{E}_\pi[\Delta\beta_{i,t}]$	0.000	(0.000)	-0.00
× Equity Risk Premium	$\lambda = 0.06$			
≈ Change in Aggregate Investment	$\Delta I_t / I_{t-1} \times 100$	**-1.28***	0.662	

- Key moment is **Covariance (investment elasticity, $\Delta\beta$)**
  - $\beta \nearrow$ affects many firms with high investment sensitivity
  - $\beta \searrow$ concentrated among few very large but inelastic firms

Cumulative sum across ranks · General equilibrium model · Details

back


(a) Market Capitalization Weights ω
(b) Investment Share Weights π
Cumulative Change in Investment
1.4%
1.0%
0.6%
0.2%
-0.2%
-0.6%
-1.0%
-1.4%
Σ = 0.0%
Σ = -0.65%
Σ = -0.99%
Σ = -1.28%
Uniform Semi-Elasticity (ϵ = -5)
Empirical Semi-Elasticity
1
500
1000
1500
2000
2500
3000
3500
4000
Market Capitalization Rank
Market Capitalization Rank

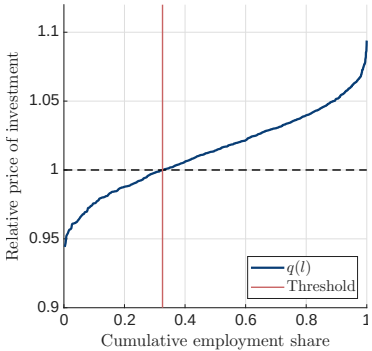
Heterogeneous firm GE model with realistic adjustment costs

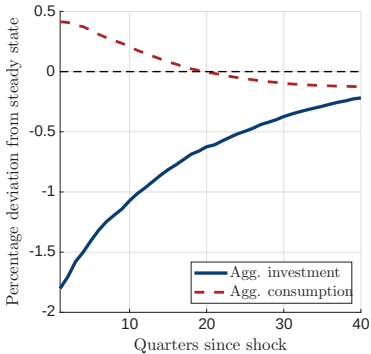
back

$$v(\varepsilon,k,\xi;\boldsymbol{s}) = \Pi(\varepsilon,k;\boldsymbol{s}) + \max_i \Big\{ -q(\boldsymbol{s})i - \frac{\varphi}{2}(i/k)^2 k - \xi w(\boldsymbol{s})\mathbb{1}\{i\neq 0\} + \mathbb{E}\left[\Lambda(z';\boldsymbol{s})v'(\cdot)|\varepsilon,k,\xi,\boldsymbol{s}\right] \Big\}$$

$$\text{where } q(\boldsymbol{s}) = 1 + \theta \times \left[\beta_0 + \beta_1 \exp\left(-a\, n(\varepsilon,k;\boldsymbol{s})\right)\right], \quad \theta' = \rho_\theta \theta + \sigma_\theta \varepsilon_\theta$$

- Calibrate adjustment cost parameters φ, ξ to Compustat
- Calibrate a, β_0, β_1 to employment share $n(\varepsilon,k;\boldsymbol{s})$ of Top-50 firms; persistence of shock ρ_θ to CAPM $\hat{\beta}$





Data on exec. compensation contracts from ISS Incentive lab

back

- Define $1(\text{Capital Efficiency Target})$ as firms which have capital efficiency targets, i.e.
  - ROIC
  - ROCE
  - ROE
  - Economic Value Added (EVA)
- Subsample of around 1,000 observations have ROIC targets available

Risk measures and uncertainty shocks

back

	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent variable: $CAPX_{t+1}$ / PPE_t (in %)					
	Measures of **?**		Measures of **?**			
Risk Exposure Measure:	Risk	Sentiment	EPU	Oil	Euro	Yen
BMI (in %)	-0.087***	-0.108***	-0.077***	-0.070***	-0.067***	-0.079***
	(0.026)	(0.025)	(0.021)	(0.022)	(0.020)	(0.020)
$\mathbb{1}\{\text{High}\}$	0.281	-0.434	-0.497	0.145	0.536	-0.548
	(0.553)	(0.700)	(0.394)	(0.536)	(0.427)	(0.391)
$\mathbb{1}\{\text{High}\} \times$ BMI (in %)	-0.027	0.036	0.010	-0.016	-0.022	0.016
	(0.029)	(0.028)	(0.019)	(0.021)	(0.023)	(0.020)
Other Controls	✓	✓	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓	✓	✓
Industry × Year × Rank-Bin FE	✓	✓	✓	✓	✓	✓
Adj. R^2	0.65	0.65	0.64	0.64	0.64	0.64
Observations	20,636	20,636	27,701	27,701	27,701	27,701

NB: Similar results using continuous variables rather than tercile indicators

Common ownership

back

	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent variable: $CAPX_{t+1}$ / PPE_t (in %)					
Common Ownership/Competition Measure:	Density	MHHI Δ	MHHI	C	Mark-Up	PCM
BMI (in %)	-0.100***	-0.098***	-0.098***	-0.107***	-0.075***	-0.089***
	(0.030)	(0.028)	(0.028)	(0.029)	(0.021)	(0.023)
$\mathbb{1}\{High\}$=1	-0.312	0.555	0.483	-0.281	1.596*	0.625
	(0.727)	(0.459)	(0.464)	(0.350)	(0.778)	(0.726)
$\mathbb{1}\{High\}$=1 × BMI (in %)	-0.007	-0.014	-0.012	0.018	-0.055	-0.029
	(0.035)	(0.024)	(0.024)	(0.021)	(0.039)	(0.024)
Other Controls	✓	✓	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓	✓	✓
Industry × Year × Rank-Bin FE	✓	✓	✓	✓	✓	✓
Adj. R^2	0.54	0.54	0.54	0.54	0.54	0.54
Observations	22,744	22,744	22,744	22,744	25,628	35,757

NB: Similar results using continuous variables rather than tercile indicators

Stocks with high-BMI are less sensitive to Tobin's Q

◂ Alt. Expl. ◂ Inv. Panel

	$CAPX_{t+1}/AT_t$		$CAPX_{t+1}/PPEGT_t$		$(XRD+0.3\ SGA)_{t+1}/K_t^{INT}$	
BMI (in %)	-0.016***	-0.015***	-0.095***	-0.091***	-0.071***	-0.072***
	(0.004)	(0.004)	(0.016)	(0.015)	(0.013)	(0.013)
Tobin's Q	0.777***	0.769***	2.863***	2.822***	2.804***	2.767***
	(0.036)	(0.036)	(0.152)	(0.151)	(0.158)	(0.153)
BMI (in %) × Tobin's Q		-0.008***		-0.034***		-0.048***
		(0.003)		(0.011)		(0.011)
Cashflow	-0.001***	-0.001***	-0.001	-0.001	0.001	0.001
	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)
Return on Assets	2.971***	3.063***	6.955***	7.255***	4.700***	5.259***
	(0.321)	(0.320)	(1.259)	(1.261)	(1.118)	(1.121)
Leverage	-0.006***	-0.006***	0.010**	0.010**	-0.024***	-0.024***
	(0.001)	(0.001)	(0.005)	(0.005)	(0.004)	(0.004)
Sales Growth	0.277***	0.272***	1.956***	1.932***	2.521***	2.446***
	(0.060)	(0.059)	(0.283)	(0.282)	(0.305)	(0.302)
Age	-0.033	-0.034	-0.240*	-0.245*	-0.092	-0.101
	(0.025)	(0.025)	(0.134)	(0.134)	(0.102)	(0.104)
Firm FE	✓	✓	✓	✓	✓	✓
Year× Ind.× Size FE	✓	✓	✓	✓	✓	✓
Adj. R^2	0.71	0.71	0.45	0.45	0.67	0.67
Observations	49,836	49,836	43,178	43,178	37,632	37,632

Stock with high BMI incorporate less private Information

back

Dependent variable:	(1) CAPX$_{t+1}$ / PPE$_t$ (in %)	(2) CAPX$_{t+1}$ / PPE$_t$ (in %)	(3) CAPX$_{t+1}$ / PPE$_t$ (in %)	(4) Price Nonsync. (in %)	(5) Bid-Ask (in %)
BMI (in %)	-0.110***	-0.116***	-0.107***	-0.150***	-0.002*
	(0.025)	(0.026)	(0.025)	(0.025)	(0.001)
Tobin's q^{tot}	2.828***	2.686***	2.605***		
	(0.180)	(0.186)	(0.176)		
$\mathbb{1}${High Price Nonsync.}		-0.360			
		(0.253)			
$\mathbb{1}${High Price Nonsync.} × Tobin's q^{tot}		0.544***			
		(0.181)			
$\mathbb{1}${High Bid-Ask Spread}			-0.026		
			(0.150)		
$\mathbb{1}${High Bid-Ask Spread} × Tobin's q^{tot}			0.500***		
			(0.148)		
Other Controls	✓	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓	✓
Year × Industry × Rank-Bin FE	✓	✓	✓		
Year FE				✓	✓
Adj. R^2	0.59	0.59	0.59	0.75	0.52
Observations	34,627	34,627	34,184	35,781	35,087

NB: Similar results using continuous variables rather than tercile indicators

No increase in probability of facing activist investor

back

Outcome y	ΔPr(y) when ΔBMI=10 p.p.	SE	p-value	No. Obs.	Mean y	# y=1
Activist Campaign (12 mos.)	-0.0017	0.0007	0.02	18,059	0.0156	282
Activist Campaign (24 mos.)	-0.0000	0.0008	0.95	18,059	0.0246	444
Activist Campaign (36 mos.)	-0.0004	0.0015	0.76	18,059	0.0328	592
Activist Campaign (48 mos.)	0.0005	0.0018	0.76	18,728	0.0408	765
Activist Campaign (60 mos.)	0.0003	0.0020	0.86	18,728	0.0479	898